

THE INVISIBLE WEAPON

*Telecommunications and
International Politics
1851–1945*

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Telecommunications and International Relations

When the telegraph was introduced in Australia in 1853, the Melbourne *Argus* went into an ecstasy of rhetoric: "We call the electric telegraph the most perfect invention of modern times . . . as anything more perfect than this is scarcely conceivable, and we really begin to wonder what will be left for the next generation, upon which to expend the restless energies of the human mind."¹ Such raptures, not uncommon in the nineteenth century, expressed not only the admiration of progressives for the marvels of technology that gave humans such power over nature, but also the optimistic faith that this power was a blessing for mankind. Thirty years later Ferdinand de Lesseps, builder of the Suez Canal, put it in these words: "All these enterprises of universal interest—some already completed, others under construction or projected—have an identical goal: drawing peoples together, and thereby bringing about an era in which men, by knowing one another, will finally cease fighting."²

The enterprises to which de Lesseps was referring were the new means of transportation and communication—the railways, steamships, canals, and telegraphs—that thrilled so many in the nineteenth century. Had he lived to see the inventions of the next century, he might have been just as amazed, but he would surely not have thought that they could make people finally cease fighting. If we have learned anything in the past century, it is that technology confers power, but that the consequences of that power are anything but predictable.

Characteristics of Electrical Communications

The relations between communication and politics are complex, and to make sense of them we first need to identify some of the relevant characteristics of communica-

tions systems. Like the mails, point-to-point telecommunications systems convey information to specific recipients. In addition, they have five other important qualities. The first four of these—speed, coverage, reliability, and cost—are technical and economic, while the fifth, security, is political and organizational in nature.

Speed is the most striking characteristic of telecommunications. What triggered the explosion of demand for the new technology was the incredible speed of electricity and its inherent promise of "real-time" information. In 1901 Captain George Squier of the U.S. Army Signal Corps expressed the astonishment of his age when he wrote: "The fastest mail express, or the swiftest ocean ship, are as naught compared with the velocity of the electrical impulse which practically annihilates any terrestrial dimension."³ Though much faster than the mail, the early telegraphs were by no means instantaneous in relaying information, but were slowed down by both technical and organizational delays. Because so much information quickly becomes stale, governments, businesses, and newspapers encouraged improvements in the efficiency of telecommunications.

Coverage refers to the number of participants in a network. The telegraph was not an isolated device, but appeared from the very beginning as a network, that is to say a complex sociotechnical system spread over a large geographic area. This was true of many other innovations of the recent past: railroads, aviation, and electric power, among others.⁴ But whereas railroads and electric power reached their optimum efficiencies on a regional scale, telegraphs quickly became national and, in the case of submarine cables, intercontinental and even global.

Telecommunications networks tend to expand for two reasons. They meet a human need and therefore attract customers like any other desirable new product. Moreover, the number of possible connections in a network—a rough measure of its value to its customers—increases as the square of the number of customers. For these reasons, telecommunications networks spread farther than their creators expected. Within twenty years of its invention, the telegraph covered Europe and North America, reaching the smallest towns. Soon thereafter, regional networks joined to form continental ones, which submarine cables then linked into a global network. Recently, as the richer countries approach the goal of a telephone in every home and on every office desk, the public, far from satisfied, demands telephones in every room, car, and airplane, and will soon want one on every wrist, capable of reaching anyone in the world at any time.

The third characteristic of electrical communication—its cost—balanced the advantage of speed and restricted its spread. Like all economic phenomena, the communication revolution touched different social groups at different moments in history. At first, long-distance telegraph rates were staggering; on the Atlantic telegraph cable of 1866, a twenty-word telegram cost one hundred dollars, or two to four months' wages for an industrial worker. For many years, the public that applauded the technical miracle of telecommunication was restrained by its cost, and only sent telegrams on the most momentous occasions.

All new technologies start out expensive and gradually become cheaper. What is amazing is that the cost of communicating (unlike, say, transportation or housing) has continued to decrease for over a century, and the decline shows no sign of abating. The public's fascination with every advance in communications technology from the telegraph on is now being rewarded at last.

Businessmen were quicker to use the telegraph, for communication offered opportunities for profit and expansion. To the business world, the speed and reliability of telegraphy often justified its expense. Numerous businesses that arose in the nineteenth century, such as railroads, shipping lines, mail-order houses, and newspapers, were intimately connected with, even dependent on, the telegraph. Soon after its introduction, the telephone also became an essential tool of business. Similarly, the radio became indispensable to shipping and airlines and created the broadcast media. Thanks to telecommunications, local businesses became regional, then national, and finally multinational, developing the world economy.

What the public wanted but could barely afford, and what businesses could afford only within limits, governments craved at almost any price. When electrical communications began, governments did more than welcome them; they often supported them and subsidized their extension, regardless of the cost.

The fourth characteristic of telecommunications, their reliability, did not come easily. The early telegraphs suffered from numerous problems ranging from equipment breakdowns to human error. Messages often got lost, delayed, or garbled. To cure the telegraphs of their teething pains required many expensive investments such as duplicate lines, better equipment, and more experienced operators. The same was true for the radio in the early twentieth century. From a technical perspective, the history of telecommunications is one of constant gradual improvements periodically interspersed with astonishing advances.

Electrical communications has often been described as one of the great achievements of mankind. But when we look at it from a security point of view, we see an entirely different picture, for security is not a technical but a social and political characteristic. Since politics have not improved since the nineteenth century but have at times gotten noticeably worse, even the history of telecommunications has a dark side.

In offering to convey personal messages from one individual to another, telecommunications systems imply that no one else will see the message. For individuals this is a promise of privacy. Businesses value confidentiality, but they also value priority of access to information. The needs of governments are even more stringent: they want many of their communications to be secret.

At the same time, it is governments that threaten the security of communications. They alone have the means to impose surveillance on their citizens and to spy on other nations. While governments have engaged in prying since the dawn of history, the ubiquity of electrical communications and the tremendous increase in the flow of vital information have made communications more vulnerable than ever. Within a country, security is a matter of faith in the discretion of government rather than a characteristic of the telecommunications system itself. Internationally, and sometimes within a country as well, communications are vulnerable to the espionage of other governments.

Telecommunications and World History

Governments crave communication, for a timely flow of information is a vital instrument of power. Even the earliest rulers needed to know what was happening in

their provinces and on their frontiers. In turn, they issued commands to their agents and subjects, hoping thereby to control events. In their foreign relations, governments were curious about other countries and eager to spread their influence, if not their raw power. And in wartime, information and control could determine victory or defeat.

Information has three levels of value: in itself, as knowledge; if it is timely, as news; and if it is exclusive, as secrets. Governments wanted all three. Large empires went to great lengths to speed the flow of information: the Romans built roads, the Persians and Mongols established relays of horses, the British subsidized mail steamers.⁵ Before the electric telegraph, however, the slow and unreliable nature of communications acted as a restraint on governmental control. In large empires it could take months for a letter writer to receive a response. In spite of their poor communications, some empires—the Chinese, the Roman, and the Spanish, for example—survived for centuries, much longer than the creations of the “new imperialism” at the end of the nineteenth century.

By the eighteenth century, European governments were no longer content to establish their own communication networks; they also accelerated the advance of communications technologies. The development of semaphores, naval flags, and public postal systems was a sign that demand was pushing against the supply of communications media.

This book covers the period from 1851, the year the first telegraph cable linked France and Britain, until the end of World War II. From the perspective of international relations, this stretch of time divides into two distinct periods: forty years of relative peace among the great powers, followed by half a century of tensions and world wars.

In the 1840s, when the electric telegraph appeared, Europe and America were in the midst of an era of peace. Old enmities had been forgotten, and military requirements had given way to the demands of civilians. The rapid spread of the telegraph responded to the needs of businesses and railways as much as to those of governments. The public and the press hailed the magic of instantaneous electrical communication. What was so important to trade and so useful to diplomacy, many believed, could only lead to a greater harmony between the peoples and nations of the world.

Gradually, telecommunications turned from a miracle into another modern convenience, a proof of the progress of industrial civilization. As the new medium spread around the world, small networks merged into large ones, while technical improvements and economies of scale led to natural monopolies. Internal telegraph networks were built and operated by the government in every country but the United States, which tolerated giant corporations like AT&T and Western Union.

The late nineteenth century was an exceptional period in two respects. First, the great powers mostly avoided fighting one another, and turned their energies to trade and imperialism in Africa and Asia. Peace made them less sensitive to security concerns than ever before, and therefore less eager to spend vast amounts of money for better communications. Meanwhile, international trade expanded so much that economics, not politics, became the motor of expanded communications. After 1860 the cable business satisfied both its investors and its customers. Even govern-

ments that monopolized their domestic telecommunications left international telegraphy to private enterprise. As we shall see in Chapters 2 and 3, the history of telecommunications in the late nineteenth century is more a history of business than of politics.

As telegraph cables reached around the world, a small number of firms, most of them British, gained a stranglehold on international communications. This dominance aroused the admiration and envy of foreigners, but little hostility. In an age of free trade nations tolerated each other's comparative advantages. As long as the European powers were preoccupied with Continental problems and the United States with its own westward expansion, the seas were Britain's special sphere. Government involvement was mainly evident in the spread of telegraphy to the colonial empires and to non-Western countries like China, as we shall see in Chapter 4. The benefits of the telegraph for colonial administration, news, and world trade were so evident that the ownership of the cables was seen as a minor issue.

Throughout much of the nineteenth century, the rivalries of the European powers had been tempered by conservative restraint and respect for the balance of power, while the territorial ambitions of their more pugnacious elements were diverted to the vast unconquered lands of Africa and Asia. As the century neared its end, however, nationalism grew more virulent while unclaimed parts of Africa and Asia became scarce. What had been, for decades, a friendly rivalry between European powers engaged in the common enterprise of imperial expansion turned to jealousy, resentment, and suspicion. The events of the turn of the century—the Spanish-American War, the Franco-British confrontation at Fashoda, the Boer War, and the Anglo-German naval arms race—exacerbated tensions and transformed the long nineteenth-century peace into the first of the twentieth-century prewar eras.

As the reliance of nations on their communications increased, so did their fears of losing them. Cable cutting in the Spanish-American War and censorship in the Boer War showed the world how vulnerable were the lines of communication, and how much more vulnerable were the nations whose global communications depended on the goodwill of Great Britain. In reaction, France and Germany hastened to build their own networks, with heavy government subsidies, and the British government began rounding out its network with subsidized strategic lines. Only the United States remained satisfied with pure free enterprise. By 1914, instantaneous communications, far from smoothing out misunderstandings between the powers, only sharpened the jitters of nervous governments teetering on the brink of war. In the hopes of freeing themselves from their dependence on foreign cables, all the great powers helped push the infant technology of radio into an early adolescence. Telecommunications, once hailed as a miracle and later regarded as a public utility, had become a political tool in the rivalry between great powers. This will be the theme of Chapters 5 to 7.

The two world wars accelerated the politicizing of telecommunications that had begun at the end of the nineteenth century. In addition to the news and private messages, governments and the military used three special forms of information: propaganda, that which governments wanted the world to believe, whether it was true or not; secrets, or the information they tried to keep from unfriendly eyes; and intelligence, or secrets that unfriendly eyes had seen. During wars and cold wars,

plain information tended to be overshadowed by these three hybrids. While every belligerent used secrecy and intelligence, control over information was especially critical for weaker nations on the defensive, for whom the smallest scrap of power could tip the balance between survival and disaster. It was in World War I that the military potential of electrical communication first became obvious, as we shall see in Chapters 8 and 9. Britain, the country with the best global communications, learned the lesson best of all.

The romance of espionage lingers on in literature long after it has been supplanted in the real world; after all, how can one compare the tedium of listening, hour after hour, to the faint crackling of a radio set with the excitement of Mata Hari's boudoir escapades? Unfortunately for literature, what the telegraph and radio did when they separated communication from transportation was to create a new form of espionage, one that was less thrilling but more powerful than any previous method of obtaining secret information. The telegraph and the radio enabled belligerents to control whole armies and fleets and to fight world wars instead of regional ones. In the process they also created communications intelligence to spy on each other. As information became easier and cheaper to transmit over long distances, governments found it more and more vital to control information by means of censorship and by secret interception, decryption, and analysis. Like so many other activities, espionage became first bureaucratic, then mechanized.

The interwar years, described in Chapters 10 and 11, did not witness a return to free trade but an exacerbation of rival nationalisms and state-controlled enterprises. Even in the United States, the last bastion of unfettered capitalism, the government became deeply involved in international telecommunications. The old British dominance was challenged on two levels: by the rise of American wealth and nationalism, and by the new kinds of cables and radio that threatened to make older technologies obsolete.

World War II was a replay of World War I with more advanced technologies. Once again Britain proved to be the master of communications. Not only were its own communications relatively secure, but it also succeeded in penetrating Germany's most secret codes and ciphers; the Americans did likewise with the Japanese. In this as in other fields, however, Britain paid for its security by relinquishing its place in the world to its ally and friendly rival, the United States, as we shall see in Chapters 12 to 14.

Though it has been over forty years since World War II, information and its hybrids—propaganda, intelligence, and espionage—are still part of the arsenal of international politics. de Lesseps to the contrary, it is questionable whether better communication will lead to peace and harmony among the peoples of the world.

International Telecommunications as a Field of Study

Information and the telecommunications systems that convey it are so essential to the modern world, it is not surprising that they have been studied from many points of view and by many different disciplines, from political science and economics to engineering and military science. A new discipline called *communication* has even

arisen in the past few years specifically to study the interactions of the new media with society.

Of the two forms of communication, broadcast and point-to-point, the broadcast or mass media have captured most of the attention of communication scholars, for two reasons: first because, by their nature, they generate publicity and seek wide audiences; and second because they create their own information and preserve it for posterity.⁶ The point-to-point media are by nature more discreet. They promise privacy, they only convey the information their customers bring to them, and they rarely retain copies of the messages. Yet they are more relevant to the conduct of government, to the outcome of wars, to the pursuit of business, even to the lives of ordinary people; who, indeed, does not value a personal letter or telephone call over a newspaper or television program? Because the messages of point-to-point media are so elusive, studies have concentrated on their structures, their origins and public history, the devices and techniques they use, and their business activities. With a few notable exceptions, the cultural aspects of point-to-point communications have yet to be studied.⁷ The political impacts of telecommunications are so crucial that they have not escaped the notice of scholars, but, because the contents of the messages are preserved not by the media but by customers who keep them secret for many years, this awareness has mainly been felt among those scholars who can afford to wait, namely historians.

In the relations between nations, the advances in communications in the past century and a half have had a dual history. From a technical point of view, it has been an era of almost uninterrupted progress, resulting in falling costs and increasing coverage, speed, and reliability. In terms of security, on the other hand, as the world has become more dangerous, the political importance of international communications has turned it into a weapon in the rivalries of the great powers.

Like any work that deals with the social aspects of technology, this one also confronts the vexed issue of autonomy. For over a century, philosophers and writers have wondered whether technology has acquired a life of its own, independent of the will of its human creators, and whether this autonomous technology is the determining factor, the "independent variable" in modern history.⁸ This question impinges strongly on our topic, since telecommunications technologies are complex and fast-changing and play an important part in the political life of nations. Yet they do not spring out of nowhere. While their origin is seldom predictable, their implementation and diffusion result neither from pure technical necessity nor from the rational choices of their inventors or their users. Rather, organizations mediate between the machines and society, influencing both. These organizations—the Eastern and Associated Telegraph Companies, the French Chamber of Deputies, the Western Union Company, the German Wehrmacht, to name a few—in effect control the flow of interactions between technology and society by purchasing, investing, subsidizing, patenting, sharing or withholding secrets, and many other means. While it often seems that technology influences the course of history—and this book is filled with examples thereof—it is in fact organizations that control the interactions between "technology" and "man" and, through their interactions with each other, make history. This book, then, is not a search for causes and consequences, but an inquiry into the interactions between telecommunications technolo-

72. African Direct Telegraph Co. Ltd., West African Cables, 1886, in Cable and Wireless, 1980; Eastern and South African TCL, "Correspondence relating to West Coast of Africa cables no. 4, 1886 & 1889," *ibid.*, 2195.

73. "Africa Agreements," Cable and Wireless, 728.

74. "Correspondence relating to West coast of Africa cables no. 4 (ex. 334), 1886 and 1889," Cable and Wireless, 2195.

75. "Prospectus of the African Transcontinental Telegraph Company Limited," Cable and Wireless, 1741; Barty-King, 86–90; Weinthal, 1:215–17.

76. "Télégraphes 1888–98," ANSOM, Série géographique, Gabon-Congo, XII, dossier 22; "Câble de Loango à Libreville," *ibid.*, dossiers 23a and 24b; and "Ligne télégraphique de Loango à Brazzaville" (1899), *ibid.*, dossier 24b; Captain P. Lancrenon "Les travaux de la mission télégraphique du Tchad (1910–1913)," *L'Afrique française: Bulletin du Comité de l'Afrique française* (January 1914), 34–40, and (February 1914), 55–69; Martial Merlin, "L'oeuvre des récentes missions en Afrique équatoriale française," *Bulletin de la Société de géographie commerciale de Paris* 36 (1914), 249–67; Daniel R. Headrick, "Les télécommunications en Afrique Equatoriale Française, 1886–1913," *Recherches sur l'histoire des télécommunications* 2 (December 1988), 73–86.

77. Robert V. Kubicsek, *The Administration of Imperialism: Joseph Chamberlain at the Colonial Office* (Durham, N.C., 1969), 30–32; Cain, 188–192; Blakeley, 65–66.

78. Robert Kubicsek, private communication, December 1989. I am very grateful to Professor Kubicsek for this information.

79. Blakeley, 66.

80. Cornelius W. de Kiewiet, *The Imperial Factor in South Africa: A Study in Politics and Economics* (Cambridge, 1937), 293.

81. John W. Cell, *British Colonial Administration in the Mid-19th Century: The Policy-Making Process* (New Haven, Conn., 1970), 43.

82. C. D. Cowan, *Nineteenth-Century Malaya: The Origins of British Control* (London, 1961), 266; Sir Andrew Clarke, *Life of Lieut.-General the Hon. Sir Andrew Clarke, G.C.M.G., C.B., C.I.E.*, ed. R. H. Vetch (New York, 1905), 155–56; Cain, 193–94.

83. Blakeley, 67.

84. Kubicsek, 109; for a similar opinion, see P. M. Kennedy, "Imperial Cable Communications and Strategy, 1870–1914," *English Historical Review* 86 (1971), 751.

85. K. S. Inglis, "The Imperial Connection: Telegraphic Communication between England and Australia, 1872–1902," in *Australia and Britain: Studies in a Changing Relationship*, ed. A. F. Madden and W. H. Morris-Jones (London, 1980), 21–38.

86. Letter of March 24, 1894, in ANSOM, Affaires Politiques, 2554/4.

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Crisis at the Turn of the Century, 1895–1901

In the early days of cables it seemed miraculous to send a message over thousands of kilometers in a matter of hours. Pulpits, podiums, and editorial pages resounded with paeans of praise for the "annihilation of time and space." By the 1880s, when cables had reached every continent, telegraphy was no longer celebrated as a miracle but had become a daily necessity of business and governments in an age of expanding global trade.

In Europe and America, the years from 1871 to 1898 were a time of peace. In the tropics, to be sure, Europeans quickly and almost effortlessly conquered whole empires. But the world was huge, there was so much land still to be taken, and diplomats were able to resolve the differences that arose between the great powers. In the last years of the century, the peoples of the Western world could admire the new technologies they had brought forth and see in them a great increase in their power over nature, ignoring, for the last time, the power these technologies gave some men over others.

This naive euphoria did not last. In the 1890s, when the advancing Western empires swallowed the last unconquered lands in the tropics, the collisions of their far-flung agents in the heart of Africa and in the expanses of the Pacific reverberated back to Europe and America, inflaming the press and agitating the chanceries. From a miracle for all mankind, the telegraph became a means of political power, a dangerous weapon in the hands of potential enemies.

Telegraphy and Diplomacy

The relationship between the diplomatic establishments and the telegraphs was two-sided, for diplomacy was also affected by the telegraphs, often in unexpected ways. The ministries of foreign affairs, with their ancient and sacred rituals, entered the

age of speedy communications with reluctance. Until 1859 the British Foreign Office received and sent telegrams at the telegraph office during normal business hours. That year, it hired two clerks to accept telegrams after hours. The State Department in Washington was much slower to enter the modern age: Not until 1868 did it have a clerk send and pick up telegrams at the Western Union office thirteen blocks away.¹

Gradually, the telegraph forced itself upon the attention of the diplomats. By the 1870s, the Foreign and Colonial Offices in London had direct telegraph lines, and some high officials, like Foreign Secretary Lord Granville and Colonial Secretary Lord Derby, had lines installed to their homes or their country estates.

As the world grew more complex, the foreign offices increased their use of the telegraph proportionately. Vary Coates and Bernard Finn have collected some statistics on the State Department's use of the telegraph. By 1910 it was handling a thousand messages a month, and by the 1930s, 200 a day. Just between the State Department and the U.S. embassy in London, the number of telegrams varied from an average of 200 a year between 1866 and 1910, to 552 a year from 1910 to 1914, 15,000 a year from 1914 to 1919, 500 to 900 a year in the 1920s, and 700 to 1300 a year in the early 1930s.²

In the nineteenth century, most telegrams were purely internal and bureaucratic: travel plans, personnel and ceremonial matters, requests for information, and the like. The telegraph was also much used for lesser foreign relations issues, such as consular matters, apprehending and extraditing criminals, and reacting to epidemics and other natural disasters.³ While this was clearly beneficial to the public, it led some diplomats to deplore, in the words of British ambassador to Vienna Sir Horace Rumbold, "the telegraphic demoralization of those who formerly had to act for themselves and are now content to be at the end of the wire."⁴

The impact of the telegraph on important events was a matter of some debate. Edmond Hammond, permanent undersecretary of the Foreign Office from 1854 to 1873, complained that the telegraph tended to "make every person in a hurry, and I do not know that with our business it is very desirable that it should be so," for it tempted officials to "answer off-hand points which had much better be considered."⁵ The French historian Charles Mazade, writing in 1875, believed that the Franco-Prussian War could have been avoided if diplomats had deliberated slowly instead of reacting to the telegraph.⁶ Not only were telegrams hasty, they were also too succinct, leaving out important details, and they often arrived out of order. At the time of the *Alabama* dispute between Britain and the United States, Foreign Secretary Granville wrote to Prime Minister Gladstone: "This telegraphing work is despairing. It will be a mercy if we do not get into some confusion."⁷

Others were more sanguine. Sir John Pender, speaking before a banquet at the Imperial Institute of London on July 20, 1894, praised telegraphy for having "prevented diplomatic ruptures and consequent war, and been instrumental in promoting peace and happiness. . . . No time was allowed for the growth of bad feeling or the nursing of a grievance. The cable nipped the evil of misunderstanding leading to war in the bud."⁸

Whether they liked it or not, diplomats could not simply ignore the telegraph and carry on in their traditional custom of long delays, for the press also got its information by telegraph, the public demanded an immediate reaction, and that

reaction was then relayed by the press to the foreign governments concerned. It was still better to have diplomacy by telegraph than by public opinion.

Charles Bright, a leading expert on submarine telegraphy, summed up its impact on international relations:

If the peoples have been brought more in touch with each other, so also have their rulers and statesmen. An entirely new and much-improved method of conducting the diplomatic relations between one country and another has come into use with the telegraph wire and cable. The facility and rapidity with which one government is now enabled to know the "mind"—or, at any rate, the *professed* mind—of another, has often been the means of averting diplomatic ruptures and consequent wars during the last few decades. At first sight, the contrary result might have been anticipated;* but, on the whole, experience distinctly pronounces in favour of the pacific effects of telegraphy.

*Indeed, it cannot be denied that there *are* occasions when rapidity in the interchange of diplomatic communications may have had—aye, and may still have—the effect of producing ruptures which "a little more time to think" would have avoided.⁹

Bright published these words in 1898, at the end of a long era of peace. What he sensed darkly, but could not openly believe, is that telegraphy could just as easily contribute to the tensions leading to war as to the rational pursuit of peace, if the men in power were so inclined.

British Cable Strategy to 1898

Great Britain's cable policy in the nineteenth century aimed at encouraging and protecting its growing cable network. The first requirement was to induce cable companies to land their cables on British soil, a goal facilitated by the location of the British Isles in the Atlantic.

Great Britain, like every country, reserved the right to grant licenses to land cables on its shores. Most countries used this right to demand concessions in return, such as lower rates for official telegrams, but Britain granted landing rights to all without restrictions. This liberal policy had encouraged cable companies to land their cables in Britain, even the French and American transatlantic cables that could technically have gone directly to the Continent. This in turn contributed to Britain's primacy as the information center of the world. As Maxime de Margerie explained:

If, despite the prodigious development of Antwerp and Hamburg, London remains the great world marketplace, it is because news from overseas arrives there first. . . . In many places the London price is the only one known, being the only one transmitted by cables which have London as their origin.¹⁰

And Thomas Lenschau added:

All important commercial and political news were known in England two to three hours before the Continent, a fact that one needs only say, in order to recognize what an incredible advantage it gives English firms over their competitors in international trade.¹¹

Outside the Atlantic, Britain's main concern was the security of its communications with its empire. One issue that cropped up again and again was the protection

of telegraph cables in wartime. In 1858, in his telegram of greetings to Queen Victoria over the first Atlantic cable, President Buchanan proclaimed:

May the Atlantic telegraph, under the blessing of heaven, prove to be a bond of perpetual peace and friendship between the kindred nations, and an instrument destined by Divine Providence to diffuse religion, liberty, and law throughout the world.

In this view will not all the nations of Christendom spontaneously unite in the declaration that it shall be for ever neutral, and that its communication shall be held sacred in passing to the place of their destination, even in the midst of hostilities?¹²

The idea of politically neutralizing cables was raised by France at the Paris conference of 1865, but nothing came of it. In 1871 Cyrus Field and Samuel Morse, echoing a common American sentiment, asked the Rome conference to neutralize cables; the plenary assembly decided that this suggestion was outside its competence, but agreed to communicate it to all interested governments.¹³ A conference on the protection of submarine cables, held in Paris in 1884, succeeded in drawing up rules to protect cables against fishing trawlers and ships' anchors, but on the more troubling issue of cutting cables in wartime, Great Britain insisted on inserting an article that read: "It is understood that the stipulations of this convention shall in no wise affect the liberty of action of belligerents." Lord Lyons, British ambassador to France and delegate to the conference, explained: "Her Majesty's Government understands Article XV in this sense, that in time of war, a belligerent, a signatory of the convention, shall be free to act in regard to submarine cables, as if the convention did not exist."¹⁴ As Great Britain had by far the most cables, the most cable ships, and the most powerful navy in the world, its wishes carried weight.

Though Article 15 plainly put cables outside the law in wartime, the legal issue continued to fascinate jurists, especially Germans and Frenchmen, who wrote many learned dissertations on the question.¹⁵ There was a reason for the British position and a reason for the French and German concern, but they only became clear much later.

In the early years of cables, the British government worried about its communications with India, which were vulnerable not only to the hazards of an unreliable technology—as in the Red Sea cable fiasco of 1858—but also to the control exercised by the countries through which its lines ran: Prussia, Russia, and Persia for the Indo-European Telegraph Company; half of Europe and the Ottoman Empire for the Turkish line; and Portugal and Egypt for the first cable to India. Since there was no hope at the time for an all-British route to India, the next best solution was to have multiple routes. This was the recommendation of the Select Committee on communications with India appointed by the House of Commons in 1866.¹⁶

For that time on, multiple and all-British routes remained the two preferred means to ensure communications security. Because of the high cost of cables, it was fortunate that most strategic cables—that is to say, those serving the big British naval bases—followed the major trade routes, so that commerce made subsidies unnecessary. By the 1880s Cape Town, Singapore, Hong Kong, Halifax, Gibraltar, Malta, Alexandria, and Aden could cable to Britain. Exceptions like Mauritius and Bermuda made the strategists uncomfortable.¹⁷

In the nineteenth century, nothing stimulated British strategic thinking as much as a good Russian war scare. The scare of 1878 led to the appointment of a Royal

Commission on the Defence of British Possessions and Colonies, chaired by Lord Carnarvon. Its reports, issued in September 1881, emphasized anew the importance of cables to the trade and defense of the Empire.

Another scare, in 1885, led to some panicky decisions, such as the Admiralty's spending £85,000 for a cable from Shanghai to Korea, which later proved useless and was sold to John Pender for £15,000. But it also had more long-lasting results. At the behest of the Colonial Office, the cabinet set up a permanent Colonial Defence Committee that included representatives of the Admiralty and the Colonial, Foreign, India, and War offices.¹⁸

The Committee believed that British communications were basically safe because cables were hard to find at sea and their landing places were protected by shore guns.¹⁹ Yet it was concerned with two areas: Bermuda, which was the only large British naval base without a telegraphic connection to Britain, and the West Indies, Jamaica in particular, which could only communicate with Britain via Cuba and the United States. As early as 1877 the Admiralty had recommended a cable from Halifax to Bermuda and on to the West Indies. The suggestion was renewed by the Carnarvon Commission in 1880, and by the Admiralty and the War and Colonial Offices in 1882, but each time it was vetoed on financial grounds, for it would have little commercial value to help defray its expense.²⁰ Upon the recommendation of the Colonial Defence Committee, however, the Cabinet finally signed a contract in 1889 for the Halifax-Bermuda section, with a subsidy of £8,100 per year.²¹

Two years later, the Colonial Defence Committee appointed a temporary sub-committee under the chairmanship of Juland Danvers to look specifically at the question of cables. Its seven members, representing the five ministries concerned with imperial defense, met six times in early 1891. On March 19, 1891, they issued a "Report of a Committee Appointed to Consider the Question of Telegraphic Communication with India in Time of War."²² This report began with the following basic principles:

2. The telegraphic systems of the world are so closely interconnected that the Committee have found it impossible to deal with the question of telegraph communication with India, without taking into consideration the requirements of the Empire as a whole.
3. The ideal of telegraph communication for purposes of war would be fulfilled by cables landing on British territory only, following recognized trade routes, and not passing near the naval stations of a possible enemy.

Instead of considering each cable individually, the committee looked at communications between the vital points of the Empire in the event of a war with the two obvious potential enemies of the time, France and Russia. Unless Britain lost command of the sea—a remote possibility indeed—the committee did not think it likely that an enemy could cut British cables. Even if some cables were cut, it would not be for long, since the British Empire owned twenty-eight of the world's thirty-six cable ships. The committee also considered the possibility that Portugal might become an "unfriendly neutral," which would jeopardize the communications passing through the Eastern station at Lisbon. Some of the most important British cables landed in Portugal or its Atlantic islands (Madeira, Cape Verde, Azores), and the British government never lost confidence in its "oldest ally."²³ All in all, the report concluded that the ideal cable network would be unreasonably expensive, but that

the network that Britain already owned was almost secure enough for any foreseeable war. It did, however, recommend two strategic cables to round out the existing network: a new cable to India and Australia, either around Africa or across the Pacific; and another from East Africa to the Seychelles and Mauritius. The former, a major project, was not implemented until the Boer War. The latter, however, was both feasible and urgent.

Mauritius was an important sugar island strategically located on the sea lanes between South Africa and India. Eager to communicate by telegraph with the rest of the world, the government of the colony had approached France about jointly subsidizing a cable linking Zanzibar with Mauritius via the French colonies of Madagascar and Reunion. This plan, while logical from a commercial point of view, horrified the British strategic planners. The director of military intelligence, H. Brackenbury, wrote a confidential memorandum that Secretary of State for War Edward Stanhope forwarded to the Cabinet:

In the event of war with France, when our commerce will be obliged to go round the Cape instead of through the Canal, Mauritius becomes at once the naval outpost of India, and Seychelles the base from which operations would be undertaken by a naval and military expedition against the French coaling station of Diego Suarez, Madagascar.

Unless immediate steps are taken to get this cable laid, an agreement will be come to between the French Government, the Colony of Mauritius and the Eastern Telegraph Company, for a cable from Zanzibar via Madagascar and Réunion, to Mauritius. Such a cable would be useless to us in time of war. . . .²⁴

Soon thereafter, a War Office conference issued a melodramatic statement:

Are we not only bound to defend the Colonies, but have we no power and no moral right to impose our veto to prevent a Colony, such as Mauritius, from making arrangements with a foreign Government which would be dangerous to us, and consequently to them also, in time of war? . . . If the contingency of war with France is to be considered at all, should it not be considered in the interests of the Colony as well as in the interests of England? Ought not a Dependency to bear in mind the same strategic considerations as the Empire to which it belongs?²⁵

Their protests were heard, for two years later the Eastern and South African Telegraph Company laid a cable from Zanzibar directly to the Seychelles and Mauritius, toward which Mauritius contributed £7,000, the Seychelles £1,000, and Britain and India each £10,000.²⁶

In 1896, when Great Britain and the United States got involved in a dispute over the border between British Guiana and Venezuela, the British government realized that all its communications with the West Indies passed through American land lines and the Key West-Havana cable. It therefore hastily prolonged the Halifax-Bermuda cable to Jamaica, at a cost of £8,000 per year.²⁷

Telegraphic Delays and French Imperialism

In 1905 George Peel, a spokesman for British cable companies, wrote:

From 1870 to the present time, Germany, Spain, Portugal, Morocco, Tunis, France, Italy, Austria, Greece, Tripoli, Crete, Cyprus, Turkey, and Russia have allowed British capital

and British enterprise to connect them with our system, in complete reliance upon the proved international uprightness and fair dealing of our citizens.²⁸

Peel notwithstanding, there were complaints about the British cables even before the crisis of 1898, and a crescendo of recriminations afterward. The complaints fell into three categories: the content of the news they transmitted, the political advantage that Britain derived from its control over the world's cables, and its unfair commercial practices.

In the case of the French Antilles, the conflict had arisen between the colonial governments and the cable company; the French government was dragged in reluctantly, and Great Britain was not involved at all. In those parts of the world where the empires were still expanding, however, especially during military actions when important events happened fast, the timeliness of the news mattered a great deal. At those moments, the imperial governments were immediately involved, and even brief delays could be construed as national affronts.

From the mid-1880s on, France actively pushed on the frontiers of its empire in Africa and the Far East, leading to occasional disagreements with Great Britain, all of which were eventually resolved. Politicians and journalists were aware of the British cable monopoly and of an occasional delay in transmitting information, but their protests were rather muted. Only after 1898 did a chorus of delayed indignation arise concerning events that had taken place a decade or two before.

The first incident took place in March 1885, during an undeclared war between France and China, when Admiral Courbet was defeated at Langson on the Tonkin-Chinese border. News of the event traveled by the cables of the Eastern Extension and Eastern Companies; the British Foreign Office received the information and transmitted it to the British ambassador in Paris before the French government heard of it.²⁹

Two years later, during the debates in the Chamber of Deputies on the French Antilles cable bill, Minister of Posts and Telegraphs Granet complained:

After the Chamber voted to entrust the laying of the Tonkin cable to an English company, on more than one occasion dispatches have arrived late in the hands of the French government, because the English companies which transmitted them gave preference and priority to telegrams of news and speculation destined for London, so that the French government was informed after the fact of the most serious events.³⁰

The next incident occurred during the Siam crisis of 1893. France had been pressing on Siam from the east, hoping to take Cambodia, if not the whole country. Great Britain was worried as always about the marches of its Indian Empire, and wanted to keep Siam independent as a buffer protecting Burma. In May 1893, after some border clashes, the French sent two gunboats to Bangkok, which the Siamese fired on. The French government thereupon authorized Admiral Humann to issue an ultimatum to the Siamese, demanding a large piece of territory and other privileges. The cable containing this authorization was held up by the Eastern Telegraph Company until the British Cabinet could read it.³¹ In the end, the British tolerated the French action because they coveted some Siamese provinces in Malaya. In 1896 they came to an understanding with France that left Siam independent, but much smaller than it had been.

The following year another incident took place in a part of the world where both the French and the British were active. When the Sultan of Morocco died on June 11, 1894, the British consul in Tangier requisitioned the country's only cable for a day and a night in order to communicate with the Foreign Office. In its June 13 dispatch from Madrid, the French news agency Havas reported:

Newspapers complain that the English cable, the only one working between Tangier and Europe after the break in the Spanish cable, was seized during the whole of last night by the English minister to communicate with the Foreign Office.

The newspapers wonder what can be, under these conditions, the security of the interests of other nations if England, holding all sources of information, can thus suspend at will communications which are not its own.³²

The following year saw another incident, this one in Madagascar, which France was coveting but where Britain had commercial and religious interests. On September 30, 1895, a French force attacked and captured the capital of Tananarive. News of the event, sent via Eastern, was held up in London for three days before the French were informed.³³

In all these cases, the delays in transmitting the news to Paris did not affect the outcome of the French imperial expansion. They only revealed the hold that Great Britain had over other countries' information, and the fact that, while the British believed their behavior was characterized by "uprightness" and "fair dealing," other nations did not have the "complete reliance" that George Peel claimed they had.

Germany and the Azores Affair

All the cable incidents that occurred in the colonial empires in the nineteenth century involved France. Germany was far behind, both in its colonial expansion and in its cable enterprises. Until 1882, German communications with America went by cable under the North Sea, then crossed Britain on Post Office land lines to the Anglo-American transatlantic cable. In 1882, the Vereinigte Deutsche Telegraphengesellschaft (German Union Telegraph Company) had a cable laid to Valentia, Ireland, to connect directly with the Anglo cable. In spite of frequent breakdowns and poor service, the Germans did not begin to plan to lay their own cable across the Atlantic Ocean until the 1890s. Beyond the North Atlantic there was as yet no demand for German cables, and few complaints about the British. The only one of any significance concerned Kaiser Wilhelm's telegram of congratulations to President Kruger of the Transvaal in January 1896; according to one writer, it was leaked to the London papers, which printed it before it reached Pretoria. Other than that, when German authors wanted to back up their statements about British misdeeds in the nineteenth century, they had to choose examples from the French experience.³⁴

Yet there was one case in which German ambitions ran up against British obstruction. Though at the time it was considered a commercial issue, in retrospect it took on a sinister political coloring; such was the evolution of Anglo-German

relations between the 1890s and the early 1900s. This was the case of the German Atlantic cable project and its landing in England and in the Azores.

As German trade with North America increased in the 1880s and 1890s, so did the cable traffic. The main line of communication was the German cable to Valentia, where it connected to the Anglo-American Company's cable to America. By the early 1890s, this route was overcrowded and Germans complained that their telegrams were given a low priority by the Anglo. In 1891, therefore, the German government agreed in principle to the creation of a new line between Germany and North America. Since it was technically impossible to bridge that great a distance directly, it sought intermediate landing points where messages could be relayed. The best places seemed to be the Azores, one-third the way across the Atlantic, and Porthcurno at the western tip of Cornwall, where the Eastern cables landed.

This scheme immediately involved a lot of other parties. Portugal had long been eager for a cable to the Azores, and had given a concession for such a cable to the Telegraph Construction and Maintenance Company in February 1892. The French government managed to have the concession transferred to the Société Française des Télégraphes Sous-Marins, which hoped it would be the first step toward a cable from Europe to the West Indies, but the Chamber of Deputies refused to subsidize this cable, and the concession lapsed in April 1893. TC&M regained the concession and laid the cable for the Europe and Azores Telegraph Company, a new subsidiary of Eastern.³⁵

Eastern looked favorably upon the German project. A German cable to Porthcurno would give the company direct access to Germany and Central Europe, bypassing the British Post Office. Furthermore, the landing in the Azores would provide a direct connection between North America and the Eastern domains of South America, Africa, and Asia, avoiding the older transatlantic companies. Thus Eastern agreed to share its Azores concession with the Germans.

On August 17, 1894, the German government signed a contract with the cable manufacturers Felten und Guillaume of Cologne to lay and operate the new Atlantic cable. All that was needed was the landing license in Britain, which the German Post Office formally requested on August 20. The Anglo-American Company and its cartel-mates the Direct United States and the Western Union companies stirred up a press campaign against the project. On October 4, the General Post Office informed the Reichspostamt that its request had been denied on the basis of an 1888 agreement that all Anglo-German cables should be owned and operated by the two governments jointly.³⁶

The refusal was, in the words of the French cable expert Charles Lesage, "a major event in the telegraphic world."³⁷ The reason officially given was transparently false, since the proposed cable would have carried no Anglo-German traffic. Nor was its purpose to protect the ailing Anglo-American Company and its cartel-mates. The real reasons are found in a memorandum dated October 26, 1894, from the General Post Office to Colonial Secretary Lord Kimberley: first, the establishment of a new telegraphic center in the Azores, and second, the loss to the British Post Office of valuable transit fees.³⁸ The British government evidently hoped, by its action, to prevent the emergence of a rival cable node between the Eastern and Western Hemispheres, one through which information would flow

outside the control of Great Britain.³⁹ If it thought so, it was mistaken, for six years later Germany laid a cable to the Azores directly, one on which the British government could neither read nor delay the traffic. In 1901 Britain authorized the Commercial Cable Company to lay a cable from Nova Scotia to Ireland via the Azores, because it was obviously futile to prevent Atlantic cables from using the Azores as a stopping point. Yet Germans interpreted the refusal of 1894 not as a communications policy, but as an insult to Germany.⁴⁰

The Spanish-American War

All of the incidents recounted above were of limited importance at the time they happened, noticed only by the telegraphic industry, the financial press, and an occasional political journalist. Only in retrospect did they seem like portents of far more dangerous confrontations. It was the events of the years 1898–1902 that changed the perceptions of the press and the politicians. It was then that the world learned that control over cables meant access to information, and that control over information was another form of political and military power. This revelation was the result of three events: the Spanish-American War, the Fashoda incident, and the Boer War. Let us look at them from a communications point of view.

For forty years, many jurists and politicians believed that telegraph cables ought to be declared neutral in time of war, but their opinion clashed with that of the British government. Yet the issue had never been tested, and it was a matter of some curiosity whether in an actual war, the belligerents would respect the cables. The question was divided into several parts: would belligerents cut cables on the open sea, or only in enemy waters? those linking an enemy to a neutral? those belonging to a neutral? those joining two neutrals but carrying information of value to the enemy? The Spanish-American War was to answer these questions.

Before 1898, many cables joined Cuba to the rest of the world. Two Western Union cables connected Havana with Key West and the United States; the West India and Panama Telegraph Company had cables from Santiago to Jamaica, from where another one ran to Bermuda, Halifax, and on to Great Britain; the Société Française des Télégraphes Sous-Marins had a cable from Guantanamo to Haiti and on to South America; and finally the Cuba Submarine Telegraph Company operated a cable along the southern coast of Cuba, from Batabano to Santiago.

When the war began, the United States had two objectives: to stay in touch with its armed forces and to prevent Spain from doing likewise. To communicate with its fleet blockading Havana, the navy brought up one of the Key West–Havana cables. Later in the war, when the U.S. Army invaded Cuba near Santiago, the British refused to allow the Americans to use their cables, claiming neutrality. The U.S. government had to requisition the French-owned New York–Haiti cable that the French, to avoid problems with landing rights, had obligingly put in the name of a dummy American corporation, the United States and Hayti Cable Company.⁴¹

Opening communications with the American forces, it turned out, was easier to achieve than cutting those of the enemy. When the war began, General Greely, Chief Signal Officer of the U.S. Army, came up with an elaborate list of actions to

take vis-à-vis different cables in wartime: Cables connecting two points in enemy territory could legitimately be cut or seized; those connecting the enemy with a neutral could be cut but only in enemy waters; neutral cables near enemy territory that might carry enemy messages (e.g., the New York–Haiti cable) should be put under military censorship; and finally censorship on distant cables, like those across the Atlantic, could be left to the superintendents of the cable companies, “men of high character, whose good faith was guaranteed by the companies whose interests they likewise guarded.” While the categorization was somewhat complex, the basic principle behind the American policy was simple. In Captain Squier’s words: “In the absence of definite international law upon the many points involved, the United States was forced to take the initiative and use this powerful military weapon for the benefit of the cause of the United States.”⁴²

So much for the theory: what actually happened was not quite what Greely had expected. On April 23, 1898, the day before the United States declared war on Spain, he ordered censorship at the Key West cable station. A few days later, the Spanish in Cuba instituted military censorship at their end of the cable. Thus the cable remained open to private and commercial messages during the whole war. Elsewhere censorship failed dismally. Spain obtained information about the United States via Canada and Mexico, most of it written by American journalists; meanwhile the U.S. Navy learned the plans of the Spanish fleet from the Associate Press correspondent in Madrid, who got it from the Spanish papers.⁴³ Censorship was no more effective than it had been in the Crimean War half a century before.

To isolate Cuba from Spain, the United States would have had to cut five cables. The navy succeeded in cutting the coastal cable near Cienfuegos, but failed to find the cables to Jamaica and Haiti. Communications between Captain-General Blanco in Havana and Admiral Cervera in Santiago, at the other end of the island, were impeded though not prevented entirely. Cuba was not cut off from Spain until the Americans took Santiago.

The reasons for the failure were obvious. The British report noted that “the failure to sever the communication may have been due not to any respect for neutral property on the high seas, but to the lack of proper appliances for cutting in deep water.”⁴⁴ And Captain Squier added: “The searching for deep-sea cables in the high seas in time of war, without an adequate chart of the location of the cable, is a difficult and very doubtful operation,” and advocated that in the future, the United States acquire a “naval cable cruiser.”⁴⁵

In the Philippines, the situation was much simpler. The islands’ only link to the outside world was the Hong Kong–Manila cable laid by the Eastern Extension Company under an exclusive concession obtained from Spain a few months before the war. On May 1, Admiral Dewey proposed to Captain-General Primo de Rivera that the cable be neutralized and used by both sides. When Primo de Rivera refused, Dewey ordered the cable cut in Manila Bay, and tried to use it to telegraph to Hong Kong. Instead of cutting the cable to Hong Kong, however, the navy had cut a local cable to Cavite. When the U.S. government discovered that Spain could still communicate with the Philippines, but the United States could not, it protested to the British, who ordered the Eastern Extension Company to seal off the Hong Kong end of the cable. Thus neither side got to use it.⁴⁶

In order to contact the United States, Dewey had to send dispatch boats to Hong Kong, and send his telegrams from there. To maintain Britain's neutrality in the conflict, the company was only allowed to accept telegrams from the American consul, not from a cruiser, and it had to distinguish between "messages to direct or influence warlike operations" and "narrative of past operations . . . for general publication as news."⁴⁷ The company finally reopened cable service to Manila on August 22, ten days after the end of hostilities.

What were the consequences of all the cable cutting and censorship? In the short run probably none, since the war was so lopsided to begin with. The only telegrams that made a difference were the two sent by the Spanish Navy Minister Bermejo to Admiral Cervera at Martinique, informing him of coal supplies near at hand and permitting him to return to Spain. Cervera never got them. In Captain Squier's words, their "accidental non-delivery undoubtedly changed the whole history of the Spanish-American War," for had Cervera received them, he might have saved the Spanish fleet.⁴⁸

In a longer perspective, the Spanish-American War was much commented on, not only by writers on telegraphic questions, but also by the British War Office, for it determined the fate of cables in wartime. It proved that, in order to sever an enemy's communications, a nation needed control of the sea and the know-how to cut cables; and the only nation that could do so was Britain.

The Fashoda Incident

In the same year, 1898, another event took place that illustrated even more forcefully the power of communications. This was the confrontation on the upper Nile between a French expedition under Major Marchand and the British under General Kitchener. While far less dramatic than a war, it was potentially more serious. The Spanish-American War, after all, pitted a major power against a lesser one, and the result was a foregone conclusion, whereas at Fashoda, two great powers came close to the brink of war and then pulled back.⁴⁹

Marchand had started out from Gabon two years before, heading east toward Ethiopia. His was not just another exploring mission; the purpose of his expedition was to claim a swath of French territory across northern Africa, from Dakar on the Atlantic to Djibouti on the Red Sea. He reached the Sudanese village of Fashoda on the Nile in July 1898. An Anglo-Egyptian force under Kitchener, meanwhile, had been fighting the Sudanese, defeating them at Omdurman near Khartoum on September 7. The British also had ambitions in Africa, at the minimum, to control the whole course of the Nile in order to protect the security of Egypt—better yet, in the minds of the more ardent imperialists like Cecil Rhodes, to take all of East Africa from the Cape to Cairo.

While the ambitions were equally megalomaniacal, the means to carry them out were anything but equal. Kitchener, hearing of Europeans on the Nile, sailed up the river on four steamboats, arriving at Fashoda on September 19. Both parties claimed the area for their country, but neither intended to fire a shot, preferring to refer the issue to their respective governments. But how? At Omdurman, Kitchener had a telegraph that ran along the railway to Egypt, and from there to England by cable.

He was therefore able to send a telegram to London on September 24, which arrived the same day, and a longer report, a summary of which was sent from Cairo on the 29th. He described the Marchand expedition as being in desperate straits:

He is short of ammunition and supplies . . . he is cut off from the interior and his water transport is quite inadequate. He has no following in the country and had we been a fortnight later in crushing the Khalifa nothing could have saved him and his expedition from being annihilated by the Dervishes. Marchand quite realises the futility of all their efforts and he seems quite as anxious to return as we are to facilitate his departure.⁵⁰

This was an exaggeration, for Marchand's force was as numerous, as well fed, and as well armed as the British, although he lacked gunboats. However, Marchand had only two ways to communicate with France: by sending messengers back to the Atlantic coast, which would have taken nine months or more, or by asking the British for permission to use their telegraph. In Paris, meanwhile, the only news of Marchand came from the British. Upon receiving Kitchener's telegram, Prime Minister Salisbury had the British ambassador in Paris read it to French Foreign Minister Delcassé. On the 27th, the latter requested facilities to communicate with Fashoda.

The British would not let Marchand use the military telegraph at Omdurman. They did, however, allow one of Marchand's associates to travel to Cairo and cable from there. Thus it was not until October 22 that the French government received Marchand's report. During an entire month, the French government believed Kitchener's story about Marchand's predicament, and on this basis decided to back down. On October 25 Marchand himself went to Cairo, only to be chastised by Delcassé for his "incredible and unpardonable" conduct in apparently deserting his post.

While Delcassé sought a way to avoid war while saving the honor of France, the British government stepped up the pressure. The Home Fleet was concentrated, war orders were signaled to the Mediterranean Fleet, and the Channel Fleet was ordered to Gibraltar. At that same moment, the cable to Dakar, the main French naval base in the South Atlantic, suddenly fell silent. Fearing the worst, the governor of Senegal mobilized his native troops and armed the cannon overlooking the harbor. Five days later the cable was repaired. Was it, as Maxime de Margerie asked with a touch of irony, "a simple coincidence"?⁵¹

The French government felt it had no choice. Marchand was ordered back to Fashoda, and on December 11 his expedition left for Ethiopia and Djibouti, from where it returned to France in May 1899.

In these events, the relative strengths of the French and British expeditions counted for little. What mattered ultimately was the power of the Royal Navy. But more immediately, the British won the diplomatic confrontation because they alone could communicate with the Sudan, and they used their control over information to deceive the French.

The British Strategic Cable Report of 1898

The Fashoda crisis, while dramatic, was over quickly. It was followed by another much deeper crisis. In South Africa, relations between the British and the Af-

rikaners had been getting worse for years, and were clearly heading towards open conflict. In this crisis, Britain felt distinctly isolated. Relations with Russia were strained as usual, and so were those with France after Fashoda; furthermore, Germany, for the first time, showed a hostile face to Britain, while the United States was at best neutral. The British government and its strategic planners had to face the possibility of fighting a war alone against another major power.

Anxieties about Britain's diplomatic "splendid isolation" appeared in the press as well as in government circles. For example, in 1896 the influential *Contemporary Review* carried an article entitled "Our Telegraphic Isolation." Its author, Percy Hurd, denounced the British cable network as "an excellent fair-weather system, but it is little more," and feared that in the event of a war, British communications would be cut in Portuguese territories, in the United States, or in the Mediterranean and Red seas. As proof, he pointed to the simultaneous interruption on both east- and west-coast cables to South Africa from December 30, 1895, to January 4, 1896, right after the Jameson raid, a particularly sensitive moment in British-Afrikaner relations. It seemed, indeed, an ominous portent. His solution, which was often to be repeated in later years, was to lay new cables to Australia and to the West Indies, touching only on British territory.⁵²

In the fall of 1898, the Cabinet appointed a committee "to consider the control of communications by submarine telegraph in time of war." This committee included not only representatives of the Admiralty, the War Office, the Colonial Office, and the Post Office, but also J. Denison Pender, head of the Eastern group. Its report, dated October 22, 1898, touched on the same topics as the 1891 report, but at greater length and in a more adamant tone.⁵³ In minute detail, it prescribed what should be done to stock supplies, station cable ships, lay new cables, defend landing sites, and otherwise prepare for war. Like the 1891 committee, it recommended that strategic cables touch only on British soil, including some very long ones: from Gibraltar to South Africa, from South Africa to Australia, and from Jamaica to British Guiana. Unlike its predecessor, it emphasized the need and disregarded the cost entirely. This provoked a protest from Chancellor of the Exchequer Michael Hicks Beach, who was not represented on the committee: "I see no limit to the demands which may be initiated by a body like the Committee on the control of telegraphic communications in time of war."⁵⁴

The committee drew up various contingency plans in the event of a war with France, Russia, Japan, or the United States, but especially France. (Evidently, the chance of a war with Germany was so remote it was not even considered.) It placed no faith in the possibility of neutralizing cables, which had been advocated by the Colonial Defence Committee of 1886:

The present Committee are not aware that . . . there has been any advance towards the neutralization of cables in war. The experience of the recent war between Spain and America . . . points to the conclusion that a nation will not hesitate to cut a cable, even when belonging to the subjects of a neutral Power, if it anticipates that any military advantage will be derived from such action. . . . It seems doubtful whether an enemy would be in any way influenced in the direction of fairness and restraint by what this country might do.⁵⁵

The committee fully expected that an enemy would try to cut British cables, but, like its predecessors, it believed that Britain's best defense was the multiplicity of its cables and its ability to repair them, protected by the superiority of the Royal Navy.

Freed of the need to worry about fairness and restraint, the committee proposed:

Seeing that there are places which Great Britain would find it very desirable, for naval and military reasons, to isolate in war, especially against France, and bearing in mind the superior resources possessed by Great Britain for carrying out this nature of attack, the Committee, after full discussion, have come to the conclusion that we ought to cut an enemy's cables wherever necessary for strategic reasons.⁵⁶

It made long lists of foreign cables to be cut, and British cables interrupted, in the event of war with France, with France and Russia, with Japan, or with the United States.

The committee did not advocate harming neutrals:

The importance to Great Britain of retaining in the hands of British Companies the main cable communications of the world, is such as to render it desirable to minimise interference with the messages of neutrals and even with the private messages of the inhabitants of a hostile country in time of war.⁵⁷

But it did advocate censorship of telegrams passing through British territory, and a discreet scrutiny of messages passing through British-owned cable stations on neutral territory. As for censorship of the news,

[S]uch a policy, though it may be suited to a country like Spain, is antagonistic to the Anglo-Saxon spirit. . . . The case, however, is different as regards India, Egypt, and those Colonies which have mainly a native or foreign population, and in those a censorship, to be put into force at the discretion of the chief civil authority . . . should be instituted.⁵⁸

While all of these suggestions seem reasonable, even mild by later standards, the report contains one item that cannot fail to astonish an inhabitant of the twentieth century: a recommendation for the appointment of a Chief Censor, under whom there would serve, among others, "an expert in deciphering." It is of course absurd to criticize the committee for not anticipating what we now know, namely that in two world wars to come, the fate of Great Britain would rest on the prowess of its cryptanalysts. What is surprising, however, is that Britain still had no code breakers at a time when the other European powers, in particular France and Russia, had elaborate "black chambers" to decrypt each other's codes and ciphers. Evidently before 1914, cryptanalysis was considered "antagonistic to the Anglo-Saxon spirit."

The Boer War

By September 1899, it was obvious that a war in South Africa was imminent. There were, at the time, two cables between South Africa and Europe, one up the east coast from Durban to Delagoa Bay, Mozambique, Zanzibar, and Aden, the other from Cape Town to Mossamedes, Loanda, Sao Tomé, and on up the west coast of

Africa. On October 3, in an effort to stop communications between the Afrikaners and their friends in Europe, the War Office imposed censorship on telegrams between the Cape and Natal on the one hand, and Transvaal and Orange Free State on the other. This was a stopgap measure, since the Afrikaners could still communicate via the Portuguese colonies. On October 14, two days after the war began, the War Office imposed censorship not only at Cape Town and Durban, but also at Zanzibar and Aden, forbidding all telegrams in code or cipher except those between foreign governments and their consuls in Africa. Five days later, the postmaster general of Great Britain broadened the ban to include government telegrams, and officially notified the International Telegraph Bureau in Berne.⁵⁹

This was perfectly legal, for according to the agreement of the St. Petersburg Conference of 1875, every state had the right to suspend any message going through its territory, "subject to notification . . . to the other contracting governments." On October 25, in response to an inquiry by the German government, the ban on government code and cipher telegrams was rescinded. Then on November 17 the British government issued a new notification to the International Bureau, once again banning all code and cipher telegrams. Clearly the British did not trust other governments, although they secretly allowed Portuguese and American government cipher telegrams through.

The other powers were quick to protest. Both France and Germany complained that the British action hurt their legitimate commercial interests in southern Africa. At the request of the German ambassador, the British made an exception to their rule, allowing the German government to communicate with the governor of German East Africa in cipher; this was carried out secretly, in order to avoid antagonizing the French by seeming to favor the Germans.⁶⁰

The British government took other actions. On January 6, 1900, the Home Secretary issued a warrant to the Post Office

to produce, for the information of the Intelligence Department of the War Office, until further notice, any telegrams passing through the Central Telegraph Office [in London], which there is reason to believe are sent with the object of aiding, abetting, or assisting the South African Republic and the Orange Free State.⁶¹

It also scrutinized cables to and from the French base at Djibouti and censored telegrams in Sierra Leone.

Finally, on January 26, in response to French and German pressure, the British government announced that commercial codes could be used, provided that two copies of the relevant codebooks were deposited at the Aden cable office. This alleviated the complaints of foreign merchants who had resented having to pay the exorbitant charges for clear-language telegrams.⁶² The ban against government cipher telegrams was not lifted until June 1900 for German communications with Southwest Africa, November for French communications with Zanzibar and Madagascar, and December for Reunion. Other forms of censorship continued past the peace treaty, and were only lifted in July 1902.

What were the results of the censorship? The secret report of the War Office Intelligence Department smugly claimed:

On the whole it would appear that little secret intelligence succeeded in passing between the enemy and their agents in Europe or elsewhere. . . . This resulted, not so much from

the importance of the messages stopped by the censor, as from the mere fact that the censorship existed.⁶³

As proof, it pointed out that shipments of canned food to Delagoa Bay for sale to the Boers stopped on December 1899; after that, the Boers were reduced to eating fresh food.

Yet censorship had another consequence, unanticipated by the short-sighted intelligence officers and, more surprisingly, even in the Foreign Office memorandum of November 1900. It proved that Great Britain was not only able to interfere with the communications between France, Germany, and Portugal and their overseas possessions, but was perfectly willing to use its control of the world cables to its own advantage. Fairness and restraint were no longer part of the vocabulary of European relations as they had been for most of the nineteenth century. It was the first lesson in twentieth-century ethics.

Notes

1. Robert J. Cain, "Telegraph Cables in the British Empire 1850–1900" (Ph.D. dissertation, Duke University, 1971), 189; Vary T. Coates and Bernard Finn, *A Retrospective Technology Assessment: Submarine Telegraphy. The Transatlantic Cable of 1866* (San Francisco, 1979), 89. See also R. A. Jones, *The British Diplomatic Service, 1815–1914* (London, 1983).
2. Coates and Finn, 89.
3. Ibid., 83.
4. Sir Horace Rumbold, *Recollections of a Diplomatist* (London, 1902), 2:111–12, cited in Stephen Kern, *The Culture of Time and Space 1880–1918* (Cambridge, Mass., 1983), 274.
5. Cain, 188.
6. Charles Mazade, *La guerre en France 1870–1871* (Paris 1875), 1:37, cited in Kern, 274.
7. Cain, 188.
8. Hugh Barty-King, *Girdle Round the Earth: The Story of Cable and Wireless and its Predecessors to Mark the Group's Jubilee, 1929–1979* (London, 1979), 101.
9. Charles Bright, *Submarine Telegraphs, Their History, Construction and Working* (London, 1898), 171.
10. Maxime de Margerie, *Le réseau anglais de câbles sous-marins* (Paris, 1909), 105–6.
11. Thomas Lenschau, *Das Weltkabelnetz*, Angewandte Geographie. I. Serie. 1. Heft. 2. Auflage (Frankfurt, 1908), 9.
12. P. M. Kennedy, "Imperial Cable Communications and Strategy, 1870–1914," *English Historical Review* 86 (1971), 730.
13. Keith Clark, *International Communications: The American Attitude* (New York, 1931), 126; George A. Coddington, Jr., *The International Telecommunication Union: An Experiment in International Cooperation* (Leiden, 1952; repr. New York, 1972), 27.
14. Captain George Owen Squier (U.S. Army Signal Corps), "The Influence of Submarine Cables upon Military and Naval Supremacy," *National Geographic Magazine* 12, No. 1 (January 1901), 8; George Grafton Wilson, *Submarine Telegraphic Cables in their International Relations. Lectures Delivered at the Naval War College, August 1901* (Washington, D.C., 1901), 12–13; Clark, 156–61.

War Office, Intelligence Department, "Telegraphic Censorship during the South African War, 1899-1902. Secret" (June 1903), preface by W. G. Nicholson, D.G.M.I., in PRO, WO 33/280; and "Memorandum on the Censorship of Telegrams to and from South Africa on the Outbreak of Hostilities with the Transvaal and Orange Free State (November 1900)" in PRO, FO 83/2196, 108ff. See also Lieutenant Colonel Thomas G. Fergusson, *British Military Intelligence, 1870-1914: The Development of a Modern Intelligence Organization* (London and Frederick, MD, 1984), 215-23.

60. "Memorandum," 7-8.

61. "Telegraphic Censorship," 11.

62. In a standard business code like the *A.B.C.*, a code word represented, on the average, six clear words; in a private code book specifically compiled for a particular firm, the ratio could be 28 to 1. In other words, codes compensated for the high cost of cabling. "Inter-Departmental Committee on Cable Communications. Second Report (March 26, 1902)," PRO, Cab 18/16/5, 39-40.

63. "Telegraphic Censorship," 26.

6

The Great Powers and the Cable Crisis, 1900-1913

After the crises of the turn of the century, the attitude of the great powers towards communications changed perceptibly. No longer could anyone rely on the kindness of strangers. Every great power needed its own communications, independent of others, with those parts of the world where it had vital interests. Some succeeded better than others, as Table 6.1 shows.

The British Pacific Cable and the "All-Red" Routes

Let us now consider the major participants in the cable-laying boom of the first decade of this century: Great Britain, the United States, France, and Germany. As usual, the British were first. And their first consideration, after the shocks of the turn of the century, was to turn their cable network into an invulnerable global communications system. This took two forms: completing their network of strategic cables, and perfecting their contingency plans in preparation for war.

One very obvious gap remained in the world cable network that radiated out from Britain in 1900: the Pacific Ocean. The Pacific cable project did not originate in colonial defense committees concerned with strategic questions but among Australians and Canadians who felt isolated at the very edge of the British world. They could not do anything about it, however, without Britain's help.¹

The idea of a Pacific cable was first suggested during the cable boom of the 1870s, but no commercial firm would even consider it. It was brought up again by Canadian railroad entrepreneur Sandford Fleming at the Colonial Conference of 1887, and received the verbal endorsement of the British Post Office, the Association of Chambers of Commerce of the United Kingdom, the Imperial Federation League, and J. Henniker Heaton, member of Parliament from Canterbury and the

Table 6.1. Distribution of Cables, 1892–1908*

Nation	1892 ^a		1908 ^b		increase	
	km	percent	km	percent	km	percent
Britain	163,619	66.3	265,971	56.2	102,352	45.2
U.S.A.	38,986	15.8	92,434	19.5	53,448	23.6
France	21,859	8.9	44,543	9.4	22,684	10.0
Denmark	13,201	5.3	17,768	3.8	4,567	2.0
Germany and Netherlands	4,583	1.9	33,984	7.2	29,401	13.0
Others	4,628	1.9	18,408	3.9	13,780	6.1
Total	246,876	100.0	473,108	100.0	226,232	100.0

*Of the 29,401 km of German and Dutch cables in 1908, 5,328 belonged to a joint venture in the Pacific.

^aU.S. Hydrographic Office, Bureau of Navigation, Navy Department, *Submarine Cables* (Washington, D.C., 1892), 41–59.

^bMaxime de Margerie, *Le réseau anglais de câbles sous-marins* (Paris, 1909), 34–35.

leading Parliamentary advocate of imperial cables.² In 1891 Hawaii, then an independent kingdom, offered to help subsidize the segment to North America. The British Colonial Office and the Treasury, however, rejected the idea on financial grounds.

Two years later, at the other end of the world, the governments of Queensland and New South Wales awarded a contract to the Société Française des Télégraphes Sous-Marins to lay a cable from Bundaberg, on the Queensland coast, to the French colony of New Caledonia. Its promoters, especially Australian cable lobbyist Audley Coote, did not see it as just a short feeder to the Eastern network, but as the first link in a cable to Fiji, Samoa, Hawaii, and San Francisco.³ That aroused the opposition of the government of Victoria and of British Colonial Secretary Lord Ripon, who insisted that the cable to Canada be all-British.⁴ It was to be part of the “All-Red” route, so named after the color of the British Empire on world maps of the period.

Behind that patriotic British opposition lay another more commercial concern. In a memorandum to the Colonial Office on January 3, 1894, Sir John Pender protested that by subsidizing the New Caledonia cable, the governments of Queensland and New South Wales “were practically committing the Australasian colonies to the laying of a Pacific cable under French auspices and control.”⁵ Coote saw it in a different light:

The aim has been for the last nineteen years to break a growing and ungenerous monopoly in the telegraph world and, now that we have so far succeeded by giving the contract to the French company, the secret allies of the Eastern Extension Telegraph Company are agitating in the press, and in the minds of public men in some parts of Australia, to try and prevent the Pacific Cable being a success. It is not because the cable touches at New Caledonia, it is because it is opposing the Eastern Extension Telegraph Company.⁶

Eastern was not interested in substituting an all-British Pacific cable for the French project, but in preventing entirely a Pacific line that would break its lucrative

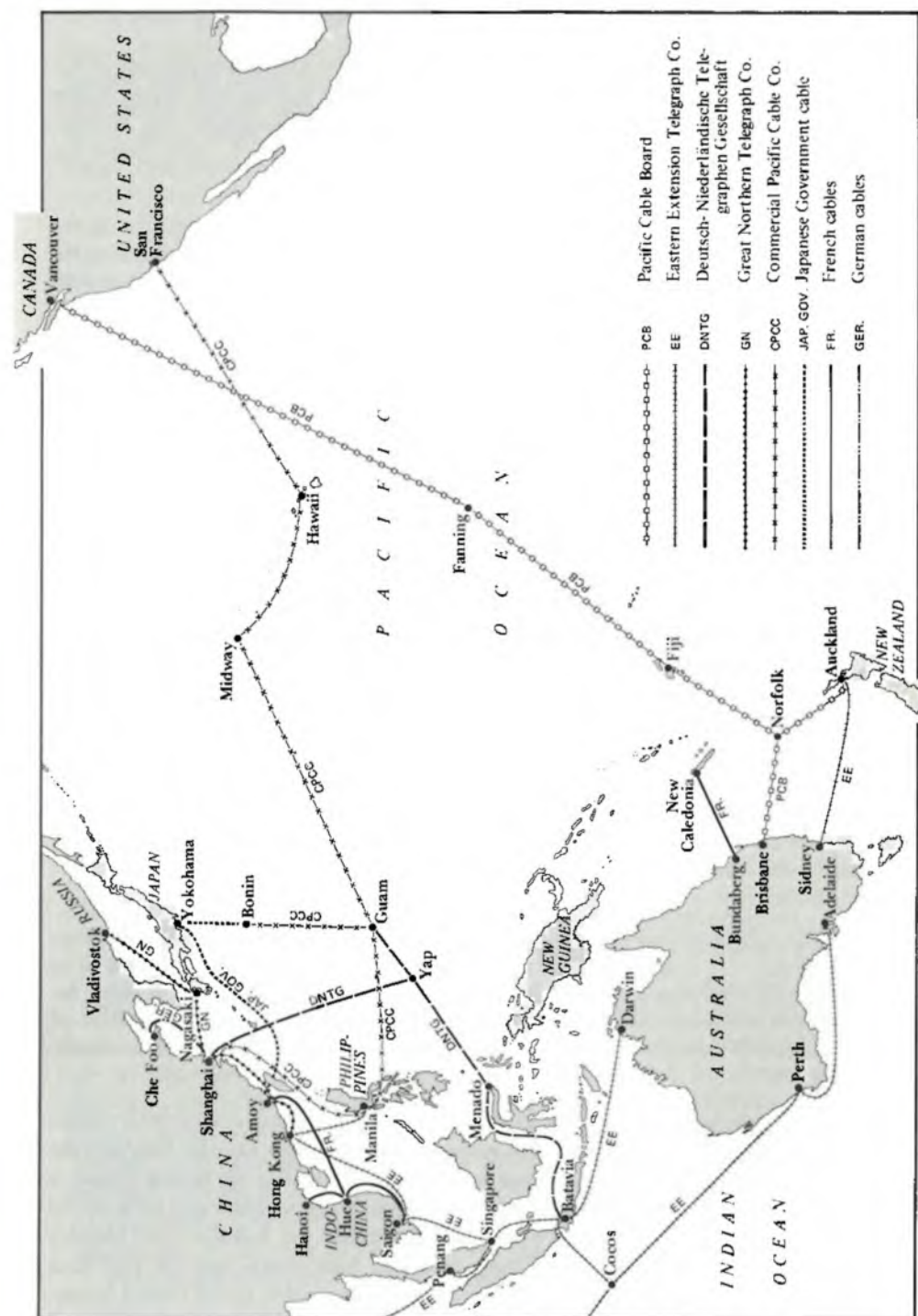


Figure 5. Pacific and East Asian Cables, c. 1905.

monopoly of the Australasian traffic. Together, the patriotic and the commercial oppositions postponed the project for another decade.

The Pacific cable issue brought out into the open the ambiguous position of the Eastern group in the British Empire. Eastern had served Britain well, laying cables to British colonies and spheres of interest, often without subsidies. The government's views were expressed in a memorandum of February 3, 1897, from the Treasury to the Colonial Office:

[Eastern] represent a real British interest, and one entitled to great consideration from the Imperial Government. They have been the pioneers of Cable communication with the most distant parts of the Empire, and it is to their enterprise and skill that the establishment of these communications is due.

In times of emergency they have always been ready to render to the Government any services which were in their power; and the advantages which have been secured from their cooperation could not in many cases have been obtained from any other body or in any other way.

It is therefore a matter of no small importance that the Government should continue to maintain friendly relations with them.⁷

The good relations were demonstrated in 1892, when R. G. W. Herbert retired as permanent undersecretary of state in the Colonial Office to become chairman of Telegraph Construction and Maintenance, or when John Denison-Pender, chairman of Eastern, was appointed to the cable committees of 1898 and 1911.⁸

But if the British government appreciated Eastern, the Australians did not. Until 1895 telegrams between Australia and Europe cost 9s 4d per word, so much that all the Australian newspapers together could only afford a few hundred words a day, and most Australians could not afford to cable "home" at all. In order to get Eastern to reduce its rate to 4s 9d, the four Australian colonies had to bribe it with a subsidy of £32,400 a year until 1899, more than the interest on a new cable across the Pacific.⁹ Henniker Heaton, himself owner of an Australian newspaper, expressed the feelings of many Australians when he spoke before the House of Commons on the Pacific Cable Bill:

I know no monopoly in the world that is doing more injury to trade than the concentrated companies represented by the Eastern Telegraph Company and its six or seven satellites. . . . I once described John Pender and Co. as an octopus which, with its tentacles in every direction, is sucking the lifeblood out of the Empire. . . . I regard the scheme now before the House as a great step forward towards the breaking up of one of the greatest monopolies the world has ever seen, and towards the consolidation of the Empire.¹⁰

During the years of procrastination, the backers of the Pacific cable never ceased trying to revive it. The colonial conference at Ottawa in 1894 strongly endorsed the project. Sandford Fleming attempted to plant the British flag on Necker Island, a speck of land at the western end of the Hawaiian archipelago that would have served as a convenient cable station between Vancouver Island and British-owned Fanning Island. But Colonial Secretary Ripon made him back down, and the Hawaiian government, which had given exclusive cable landing rights to the United States, quickly sent a ship to annex the island instead.¹¹

In 1895, the idea gained an important proponent in the new colonial secretary, Joseph Chamberlain, who advocated a Pacific cable "to communicate with one another without having to rely on the sufferance of foreign Powers."¹² In 1896 he appointed a Pacific cable committee with representatives of Canada, New South Wales, and Victoria, which recommended a state-owned Pacific cable. New South Wales, Queensland, Victoria, and New Zealand each agreed to pay one-ninth of the cost of the cable, and Canada five-eighteenths; that left five-eighteenths for Britain. In an attempt to block the project, Eastern offered to lay an all-British cable to South Africa and another from South Africa to Australia, on condition that Britain refrain from subsidizing any competitive venture.¹³ To overcome the opposition of Chancellor of the Exchequer Hicks Beach and the Eastern group, Chamberlain took the issue to the Cabinet in February 1899. Hicks Beach accepted the argument that the cable represented a quid pro quo for Australasian and Canadian troop contributions to the Boer War. Parliament passed the Pacific Cable Bill in August 1901.¹⁴

Since Eastern would not cooperate even with a subsidy, the cable was entrusted to a new body, the Pacific Cable Board, representing Britain, Canada, the Australian colonies, and New Zealand. It signed a contract with TC&M in December 1900. The undertaking was difficult. Soundings had to be taken where none had ever been taken before. The section from Vancouver Island to Fanning Island, at 6,560 kilometers, was the longest single cable span on earth, so long that a new ship, the *Colonia*, had to be built to hold it. From Fanning the cable continued to Fiji and Norfolk Island, from which two lines ran to New Zealand and Queensland, respectively. The cable was finished in October 1902 and opened to the public on December 8.¹⁵

Great Britain had finally agreed to lay a Pacific cable because the tensions at the turn of the century had overcome fiscal and commercial considerations. Ironically, the cable turned out to be good business. Its first effect was to force Eastern to lower its England-Australia rate from 4s 9d in May 1900 to 3s in January 1902, to the great satisfaction of the Australians. It was also faster: a telegram took an hour instead of a day. In the long run, it carried enough traffic and accumulated enough profit to warrant the laying of a duplicate cable in 1921-22.¹⁶ But was it all-British? Across the Pacific and Atlantic Oceans, yes, and also across Canada, where the telegraph ran alongside the Canadian Pacific Railway, "except for 270 miles when the line was diverted through the American state of Maine."¹⁷ A minor omission.

The Pacific cable was not the only strategic cable laid in the immediate aftermath of the Boer War. Indeed, the armed forces and the Colonial Defence Committee developed what P. M. Kennedy called "a virtual fetish" for all-red cables.¹⁸ Two were laid in the same years as the Pacific cable. The first of these was a new cable, completed in early 1900, from Cape Town to St. Helena, Ascension, and St. Vincent in the Cape Verde Islands; while not truly "all-red," since the Cape Verdes belonged to Portugal, it was considered much safer than the three that followed the African coasts, which were prone to natural breakdowns and vulnerable to foreign interference. A year later another cable was laid from Ascension to Sierra Leone.¹⁹

New cables were also laid across the Indian Ocean in 1901 and 1902: from Durban to Mauritius, from Mauritius to the Seychelles and Ceylon, from Mauritius to Perth via Rodrigues and Cocos islands, and from Cocos to Singapore. While

duplicating existing lines, they provided an alternative route to India, the Far East, and Australasia in the event Britain lost control of the Mediterranean.²⁰ Finally in 1901 a cable was laid, at Admiralty request, from the British naval base at Weihaiwei to Taku in support of military operations against the Boxers in north China.²¹

British Cable Strategy, 1902–1914

By 1902, Great Britain possessed not only the world's major commercial cables, but also a set of strategic cables that made its communications with its major colonies and naval bases practically invulnerable to attack.

But having cables was not enough. In a world that was heading for war, strategists made ever more elaborate contingency plans and worried about defensive and offensive actions in minute detail. The first of the committees to concern themselves with cables after the Boer War was the Inter-Departmental Committee on Cable Communications. Its first report, dated August 8, 1901, dealt with the question of telegraph rates between Britain and India. Its second report of March 26, 1902, considered all the other issues.²² It restated the basic principle behind British cable policy:

We regard it as desirable that every important colony or naval base should possess one cable to this country which touches only on British territory or on the territory of some friendly neutral. We think that, after this, there should be as many alternative cables as possible, but that these should be allowed to follow the normal routes suggested by commercial considerations.²³

It also repeated many of the recommendations of previous cable committees—the importance of completing the British network in the Indian Ocean, the contribution of the Eastern group to British security, and the need to maintain the profits of the private companies. In essence, the committee was satisfied with the security of British cable communications.

The strategists could now turn to offensive operations. In August 1904 the Admiralty issued a "Memorandum on the Protection of British Submarine Cables, and on the Destruction of an Enemy's Cables," which was mainly concerned with a possible war against France, with or without Russia.²⁴ In 1904 the War Office prepared a document entitled "Censorship of Submarine Cables in Time of War." Noting that the International Telegraph Convention of St. Petersburg required that other governments be notified before any telegraph service was suspended "except where the advice might be dangerous to the security of the State," it recommended that "wherever there was a suspension of service under that Article it would be held dangerous for the security of the State to advise the office of origin, and therefore no advice should be sent."²⁵ In other words, the British attitude toward state security was edging closer to that of Russia and Germany.

In 1908, with the "Report of an Inter-Departmental Committee appointed to draw up instructions for establishing a secret censorship of submarine cables at certain foreign stations in time of grave emergency," Britain moved yet another step further from peace²⁶:

During a period of strained relations with a foreign Power, when war, though not inevitable, may break out at any moment, it may become necessary, at certain stations abroad, to take measures to prevent the disclosure of information regarding naval or military preparations and movements. The inauguration of the scheme of open cable censorship arranged to be put in force in time of war might in such circumstances precipitate hostilities; it is therefore necessary to resort to some form of secret censorship.

The last major report on cable strategy was "Submarine Cable Communications in Time of War," issued by the Standing Subcommittee of the Committee of Imperial Defence on December 11, 1911. On the issue of defense, the report noted with satisfaction that Britain now had an "all-red" system: "The dependence of the United Kingdom on cable stations situated upon foreign territory for the transmission of telegrams has been generally eliminated."²⁷ As to the offense, the report acknowledged for the first time the German threat and gave detailed instructions on how to cut Germany's communications without offending potential neutrals like the Netherlands and the United States. It was a much simpler task than isolating France: "Generally speaking, if France and Russia were in alliance with this country, it would be possible to isolate Germany from practically the whole world, outside Europe, by cutting the cables to the Azores, Tenerife, and Vigo and the three cable landings on Yap Island."²⁸ Finally, in another innovation, it recommended the construction of wireless stations to supplement cables at vulnerable places.²⁹

Great Britain had finally achieved the ultimate goal of communications strategy: Its own cable network was invulnerable to attack or interference, and it had the ability to spy on its enemies or isolate them from the rest of the world. It was an important weapon in the world wars to come.

The American Cables

Laying a cable across the Pacific appealed to Americans as well as to Australians and Canadians. In the early 1870s Cyrus Field, promoter of the Atlantic cables, proposed cables to Japan and Russia, but had to abandon the scheme in the face of technical and political difficulties. Several times during the 1870s, Congress discussed the proposal, but appropriated no money. The government of Hawaii was more interested: It promised exclusive concessions to Cyrus Field and later to Audley Coote, but they lapsed before they could be carried out.

In the 1880s, President Grover Cleveland supported the idea, and in the early 1890s, the Hawaiian legislature, dominated by American businessmen, granted an exclusive concession and a yearly subsidy of \$25,000 to General Hartwell. Several bills involving an American subsidy for a Pacific cable passed the Senate but were defeated in the House of Representatives.³⁰

By 1895, as trade across the Pacific increased, the cable idea attracted the attention of several capitalists. One of them was Colonel Zephaniah S. Spalding, a businessman with interests in Hawaii, America, and Europe. Supporting Spalding secretly was none other than John Pender, who had fought so hard against the all-British Pacific cable. The president of Hawaii, Sanford B. Dole, promised Spalding an exclusive twenty-year concession and \$40,000 a year on condition that the United States government also grant him a subsidy. Spalding then founded the

Pacific Cable Company of New Jersey, which was to have joined the Eastern Extension and the Great Northern companies in laying and operating the cable. The following year, the Western Union Telegraph Company became involved through J. Pierpont Morgan and James A. Scrymser, director of the Mexican and Central and South American Telegraph companies. The two men founded a rival firm, the Pacific Cable Company of New York, and attacked Spalding's company as a front for the Eastern interests. Both companies lobbied Congress for subsidies, but their mutual accusations only achieved a deadlock.³¹ The Pacific Ocean and the Far East were still very peripheral to the interests of American politicians.

Then came the Spanish-American War and the annexation of Hawaii. Quite unexpectedly, the United States found itself in possession of an empire stretching across the Pacific Ocean. A cable was suddenly a strategic necessity. In a special message to Congress on February 10, 1899, President McKinley said:

The necessity for speedy communication between the United States and all these Pacific islands has become imperative. Such communications should be established in such a way as to be wholly under the control of the United States, whether in time of peace or of war. At present the Philippines can be reached only by cables which pass through many foreign countries, and the Hawaiian Islands and Guam can only be communicated with by steamers, involving delays in each instance of at least a week. The present condition should not be allowed to continue for a moment longer than is absolutely necessary.³²

In his annual messages to Congress on December 5, 1899, and December 3, 1900, McKinley repeated his pleas. After his assassination, his successor, Theodore Roosevelt, did likewise:

I call your attention most earnestly to the crying need of a cable to Hawaii and the Philippines, to be continued from the Philippines to points in Asia. We should not defer a day longer than necessary the construction of such a cable. It is demanded not merely for commercial, but also for political and military considerations.³³

Between January 1899 and December 1901, some eighteen Pacific cable bills were introduced in Congress. Some called for government construction, ownership, and operation of the cable; others would have entrusted it to private enterprise with a government subsidy. The leading proponents of the government cable were Rear-Admiral Bradford and Captain Squier of the Signal Corps, while Scrymser and E. L. Baylies of the Pacific Cable Company of New York, among others, lobbied for private ownership.³⁴ But it was all in vain, for Congress, deadlocked between the public and the private cable proposals and not yet convinced that an empire cost money to maintain, rejected them all.³⁵

On August 22, 1901, the Pacific cable issue, which had been tossed about for twenty-five years, was suddenly resolved. John W. Mackay, president of the Commercial Cable Company and the Postal Telegraph Company, rival to Western Union, offered to lay a cable to the Philippines without a government subsidy. He also promised to lower the rate of a message from San Francisco to the Far East to a dollar a word from the \$2.35 charged by Eastern. He founded the Commercial Pacific Cable Company, bought a cable from the India Rubber Company in England, and had it sent to the Pacific. Instead of requesting a landing license, he proceeded under the Telegraph Act of 1866, which "gives the right to any telegraph

company organized under the laws of any state, and complying with its conditions, to construct and operate lines of telegraph throughout the domain of the United States, and over, under or across the navigable streams or water of the United States."³⁶ Now that Hawaii, Guam, and the Philippines were American, the Pacific Ocean had become a "navigable water of the United States." It was a wise move, since neither the executive branch nor Congress had any established procedure for granting landing licenses. The Commercial Pacific cable reached Manila on July 4, 1903. In 1906, it was connected to Japan and China.³⁷

Mackay's enemies accused him of having made a secret deal with the Eastern conglomerate, but he stoutly denied it. Twenty years later, in the Cable Landing License Hearings before the Congress, his son Clarence Mackay admitted that the Commercial Pacific Cable Company was only one-quarter owned by the Commercial Cable Company; another 25 percent belonged to the Great Northern Company, and the remaining 50 percent to the Eastern and Eastern Extension companies. In other words, his father had lied.³⁸ On June 26, 1904, seventeen agreements were signed between the Eastern group, the Great Northern, Commercial, Commercial Pacific, German-Dutch, and Indo-European Telegraph companies, and the Chinese and Russian telegraph administrations. They formed a giant cartel to apportion all telegraph traffic in the Pacific, and to ensure that the rates from China to Europe would be lower via Russia or India than across the Pacific and America.³⁹ It was all done secretly, and did not appear in the reports of any of the companies involved.

The reason for the agreements was that the Eastern Extension and the Great Northern had exclusive concessions to land cables in China and Japan, and the Eastern Extension even claimed the exclusive right to land cables in the Philippines, based on a Spanish concession that the United States had rejected.⁴⁰ In theory, it should have been impossible to lay a cable from America to the Far East without their permission. But for the Eastern and Great Northern companies to have stood on their rights would have caused a furor in the United States, and they dared not risk such a confrontation. The solution was to back an American enterprise. It had to be done in the greatest secrecy, however, to avoid offending not only American but also British sensibilities, for it would have seemed in poor taste for Eastern to support an American Pacific cable after trying for so many years to prevent a British one. In exchange for providing most of the capital and—more importantly—the cables that could only be obtained in Britain, the Eastern and Great Northern got control of the rates, and thereby secured their stranglehold over communications in the Far East and the maintenance of their profits. In the nineteenth century, when governments had been relatively peaceable and cooperative, enterprises were exclusively of one nation or another. In the twentieth, ironically, just as peoples and politicians were entering an era of strident and bellicose nationalisms, private firms were becoming more international-minded. But they had to be rather discreet about it.

In the Atlantic, as in the Pacific, the United States government stayed out of the cable business and let private enterprise operate on its own. The result was a tremendous expansion of American cables at the expense of British interests. In 1909 AT&T, already a telephone monopoly, bought the Gould family holdings in Western Union. AT&T's president, Theodore Vail, hoped to consolidate America's telephone and telegraph lines into one giant network. Since Western Union owned

some Atlantic cables, the government threatened to take the company to court on the grounds that the Atlantic cable pool violated the Sherman Anti-Trust Act. To avoid a suit, Vail pulled Western Union out of the pool. In 1911 the British members of the pool, the Anglo-American and Direct United States companies, suddenly found themselves cut off from Western Union's nationwide network of land lines and offices, and were about to lose all their business. To avoid bankruptcy, they were forced to lease their cables to Western Union. Three years later, AT&T sold its holdings in Western Union. Ironically, the government's legal threat had only created a more monopolistic situation than before.⁴¹

Internationally, the results were ambiguous. The cables were British owned but operated by an American company that employed British cable clerks. One British expert, the engineer Charles Bright, found it frightening that Great Britain no longer controlled a single Atlantic cable:

Not a single one of the Atlantic cables . . . could at any time be relied upon as a genuinely All-British strategic connection in the event of trouble with the United States or with other countries interested in disturbing our Imperial communications.⁴²

The British government, however, was not worried. If a German or French company had leased the cables instead of Western Union, it would no doubt have stepped in to prevent it. But the report of December 1911 on "Submarine Cable Communications in Time of War" did not even consider the possibility of a war with the United States. What mattered from a strategic point of view was neither ownership nor management of the cables, but their location. Since all the leased cables passed through Britain, the government could censor or scrutinize the traffic as easily as under British management. In effect, the British and American cable interests were not rivals, but close allies.

France and the Cable Crisis

In 1898, the United States was just beginning its imperialistic career outside North America, and American politicians were still rather naive about the dangers of global competition. Hence the American Pacific cable project was motivated more by financial than by political considerations. France, in contrast, already had a long history of overseas conquests and an even longer experience of European power politics. The French were thus acutely sensitive to their position in the world relative to other nations. The events of the turn of the century affected France directly and triggered a political reaction.

In certain ways, France was in an advantageous position with respect to global communications. It faced both the Atlantic and the Mediterranean; it had colonies on every continent, which provide both a demand for communications and convenient landing places for cables; and it was a rich nation, able to invest large sums abroad. In other ways, however, France was at a serious disadvantage compared with Britain. Its colonies were far poorer and, except for Algeria, none of them had white settlers. It had less industry and less overseas trade. Its politicians were preoccupied with the German threat and, with few exceptions, less interested in overseas questions. Its merchants did little trade with the French colonies, other than

Algeria. Only in the 1890s did public opinion begin to take pride in the empire. But worst of all was the fact that France had lagged behind in the first cable boom, and the British had seized all the profitable routes.

The result was visible in the fate of the French cable enterprises. The first of them, the Société du Câble Transatlantique Français (or Société Erlanger), was absorbed by the Anglo-American Company in 1873 after four years of operation. The second one, the Compagnie du Télégraphe de Paris à New York, or PQ, limped along from 1879 to 1895. Both were exclusively transatlantic, while a third firm, the Société Française des Télégraphes Sous-Marins, operated short cables with government subsidies in the West Indies and between Australia and New Caledonia. When the PQ company went bankrupt in 1895, the French government forced it to merge with the Société Française, creating the Compagnie Française des Câbles Télégraphiques, or CFCT. In contrast to British cable companies that treated the government of Britain as a valued customer and occasional collaborator, the French company was so deeply in debt to the government and so tightly controlled as to be essentially an official agency. This was to be the vehicle for France's cable policy.⁴³

Before 1898 only a few voices in France called for more cables. In 1886 and again in 1889 the Chamber of Deputies had refused to vote credits. In 1894-95 the cable lobby, which was identical to the colonial lobby, tried again. The journalist Harry Alis, head of the Union coloniale, wrote a series of articles in the *Journal des Débats* in July 1894 calling attention to the fact that in every French colonial harbor there sat a British "agent" employed by a British cable company, watching the French ships; that the British used their power to withhold vital information, as in the Tangier case; and that France was spending 2,337,000 francs (£93,480) a year subsidizing British cables instead of laying its own.⁴⁴ The Société des études coloniales et maritimes, a colonial lobby, presented the following resolution to the Chamber of Deputies with the support of the ministers of war, navy, and colonies:

Whereas it is necessary for a great country like France to have independent national telegraphic communications with its colonies, in order to avoid, on the one hand, the repetition of serious problems like those which were noticed during the Tonkin and Dahomey expeditions and the Siam affair, and, on the other, to guarantee to our overseas commerce advantages equal to those of our foreign competitors until now privileged, the Société des Etudes Coloniales et Maritimes expresses the wish that a sufficient sum be put into the budget each year to serve as a subsidy or guarantee of interest to French telegraph lines, in existence or to be created, destined to link the French colonies to the metropole.⁴⁵

At the same time, J. Depelley, head of the Compagnie Française des Câbles Télégraphiques, spoke before the Union Coloniale Française on the subject of "Submarine Cables and the Defense of Our Colonies":

The nation whose colonial interests are the most often in opposition and at war with ours is England. Is it not imprudent to leave our means of correspondence in its hands? It is already a mark of inferiority from the commercial and political point of view; it would be a danger if we had to defend our colonies.⁴⁶

But their efforts were in vain, for the Chamber of Deputies was too parsimonious and too preoccupied with the defense of France against Germany to be swayed by such arguments.

What changed the situation was the Fashoda crisis and the Boer War. On November 27, 1899, a group of deputies, led by Deputy Henrique, presented the first cable project to the Chamber. In December Colonial Minister Decrais and Commerce Minister Millerand announced to the Chamber that the Council of Ministers was now convinced of the need for a French cable network. On January 30, 1900, the government presented a bill on "the establishment of a network of submarine telegraph lines destined to connect certain French colonies to the metropole and to extend the land lines in the French colonies of West Africa." It called, in the first place, for cables from Indochina to Hong Kong, from Madagascar to Reunion, from Marseille to North Africa and Senegal, and land lines from Senegal to the other West African colonies; and after that, for cables from France to Senegal, and from Gabon to Madagascar. This bill received the support of the committee on colonial affairs and of a number of influential deputies.⁴⁷

The parliamentary debates were accompanied by a chorus of publicity by cable and colonial lobbyists. The Syndicat de la presse coloniale, the Groupe colonial des conseillers du commerce extérieur, and various business groups in Marseille and Bordeaux issued proclamations. Books and articles appeared in the press denouncing British companies, pointing out how vulnerable France would be in the event of a war with Britain, and demanding an all-French cable network.⁴⁸

Finally in July 1901 the Chamber passed two laws: one "authorizing expenditures for the creation of a submarine network," which the government used secretly to buy the South American Telegraph Company; the other "authorizing the purchase of the cables of the West African Company," which ran along the West African coast from Senegal to Gabon. Two years later the Chamber authorized the laying of cables from Brest to Dakar, from Madagascar to Reunion and Mauritius, and from Indochina to Borneo.⁴⁹

In the nineteenth century, France had acquired two cables across the Atlantic, several joining France to Algeria and Tunisia, a small network in the West Indies linked to New York since 1896, and short pieces linking Madagascar and New Caledonia to the British network. They now proceeded to expand their communications channels in two ways: wherever possible by buying or laying cables, and elsewhere by joining nations other than Britain in forming alternative routes.

The first objective was South America, where France had commercial and political interests. In 1901 the French government paid the India Rubber Company £365,000 for the shares of the South American Telegraph Company, which had operated a cable from Dakar to Pernambuco since 1892, but never paid a dividend. It was done secretly in order to avoid offending Brazil, which would not have tolerated a foreign government agency operating on its shores. The purchase created an anomalous situation: a company that was at the same time British and French, public and private. It was also a commercial mistake, since Pernambuco was a backwater, and Brazil had given an exclusive concession to the Western Telegraph Company, an affiliate of Eastern, for cables along its coast to Rio and beyond. Therefore, the French were just as subject to British power as before. In 1914 the firm was renamed Compagnie des Câbles Sud-Américains, and its headquarters were moved to Paris.⁵⁰

In February 1902, the French government made its second major purchase. For £144,000 it bought from the West African Telegraph Company those sections of its

cable that connected Dakar with Conakry (Guinea) and Grand-Bassam (Ivory Coast) with Cotonou (Dahomey) and Libreville (Gabon). Intermediate stops at non-French ports were cut out. The stretch from Conakry to Grand-Bassam was covered by a land line through the interior. Three years later France laid a cable directly from Brest to Dakar. From then on it had an independent all-French cable to its West African possessions. In 1912–13 it completed the line with a piece from Conakry to Grand-Bassam and an extension from Libreville to Pointe-Noire in the French Congo.⁵¹

More distant French colonies got less adequate service. It had always frustrated French colonialists that their richest possession, Indochina, could only communicate with France through the Eastern cables. In 1901, therefore, a cable was laid between the Vietnamese town of Tourane and Amoy in China, connecting with the Great Northern cable. Léon Mascart, the captain of the cable ship *François Arago*, which laid the cable, saw it as "a dazzling proof of the perfection of our industry," and proclaimed: "to cable from Indochina to Paris without being subject to the supervision of the English offices . . . the emancipation from English censorship. . . . Thanks to this new cable, Indochina has a route which, if not entirely French, is at least friendly and neutral."⁵²

Five years later another cable was laid between Saigon and Pontianak in Borneo, where it joined the Dutch and German lines to the American Pacific cable at Guam. Now France had two non-British means of communicating with Indochina. There were even plans to lay cables from Java to Reunion, and from Madagascar to the French Congo, either by cable around Africa, or somehow across Tanganyika and the Belgian Congo.⁵³ These grandiose schemes were totally unrealistic, as they would have required enormous subsidies and the cooperation of many other countries. Reality was also a matter of business. Both non-British routes to Indochina had to charge higher rates than those of Eastern. Hence all the business traffic continued to use Eastern, and the cables consistently lost money. When they broke a few years later, they were judged not worth repairing and were abandoned.

What then were the results of the French desire for cables after 1900? Only the cables to North America and to North Africa showed a profit; the rest were commercial failures. All in all, the French government paid between six and eight million francs (£240,000–320,000) per year for cables between 1900 and 1914.⁵⁴ But of course the cables were not laid for commercial reasons. Their ostensible purpose was strategic, more specifically because of the fear of a war with Great Britain. Although that fear evaporated when France and Britain signed the Entente Cordiale in 1904, the cable-laying momentum continued, impelled by the vested interests it had created and the continued efforts of the cable lobby.⁵⁵ This implies that the cables were not really strategic, either, for if there had been a war with Britain, they would have been cut. It is in peacetime that they served their purpose: to keep France among the great powers in the realm of global communications.

Germany and the Cable Crisis

Behind France lay Germany, a country with aspirations to greatness. A latecomer to the community of nations, it had missed the imperialist banquet and had to content

itself with leftovers, minor colonies hardly commensurate with its swelling national ambitions. From the standpoint of communications, they were too distant and too far apart to be of much use in a cable network, and too poor to warrant a commercial investment. Even in the Atlantic, where Germany had serious business interests that fully justified direct cables, its location had long frustrated any telegraphic plans. When the German government at last determined to build its own cable network after the cable-landing incident of 1893–94, it found itself at a great disadvantage vis-à-vis other nations.

German submarine telegraphy was the product of a cooperative effort between three parties: the Felten und Guillaume Carlswerk A.G., manufacturers of wire rope and cable, or more exactly their owners, the brothers Emil, Max, and Theodor Guillaume; the Reichspostamt, or Imperial Post Office, especially the directors of its telegraph department, Dr. von Stephan until 1901 and Dr. Sydow after that; and the A. Schaaffhausen'scher Bankverein, a Cologne bank. In 1896 Felten und Guillaume founded the Deutsche See-Telegraphengesellschaft, which operated a cable to Vigo in Spain, where it connected with the Eastern network.⁵⁶ Three years later, it started a cable-making subsidiary, the Norddeutsche Seekabelwerke, and bought three cable ships. The acquisition of cable technology posed no difficulty; by the 1890s Germany had the most sophisticated electrical industry in the world. The firms Siemens und Halske of Berlin and Siemens Brothers of London had long cooperated on technical matters, and many German cable experts—Wilhelm Siemens and Eugen Obach in particular—had worked in Britain. By the 1890s German shipbuilding was catching up with the British. As soon as the demand appeared, German industry was ready to supply the cables.

In the next few years, Felten und Guillaume also founded the Deutsch-Atlantische Telegraphengesellschaft, or DAT, to operate cables to North America; the Deutsch-Südamerikanische Telegraphengesellschaft for Africa and South America; the Osteuropäische Telegraphengesellschaft for Eastern Europe; and the Deutsch-Niederländische Telegraphengesellschaft for the Pacific.⁵⁷ Felten und Guillaume was thus in a good position to benefit from the sudden enthusiasm for cables that erupted in Germany in 1900.

What triggered the boom was a combination of economic demand and political indignation. The demand resulted from the growth of the Germany economy and of its overseas trade, as shown in Table 6.2. Much of this overseas traffic was transatlantic, and the number of telegrams exchanged between Germany and the United States also grew in this period, from 30,000 in 1880 to 343,292 in 1901 and 710,376 in 1910.⁵⁸ Thus, even had there been no political motives, the growth of trade would have fully justified laying cables from Germany to North America and even to South America. What politics did, in Germany as in France, was to divert cable investments to other less profitable routes, such as to Africa and in the Pacific, where German business interests were marginal at best.

As in France, it was British censorship in the Boer War that triggered the political indignation that led to a German cable policy. In November 1899 the government asked the Reichstag for credits to build a cable network. The first appropriation, an annual subsidy of 1,400,000 Reichsmark (£67,000), went to the DAT, founded by Felten und Guillaume and A. Schaaffhausen'scher Bankverein earlier that year to lay a cable across the Atlantic.⁵⁹

Table 6.2. German Commerce and Merchant Marine, 1882–1913^a

Year	Commerce outside Europe	Merchant marine
1882	499,715,000 Reichsmark	458 steamships 251,648 tons net
1894	1,944,500,000 Reichsmark	1016 steamships 823,702 tons net
1913	7,277,000,000 Reichsmark	2098 steamships 2,655,000 tons net

^aCharles Lesage, *La rivalité anglo-germanique. Les câbles sous-marins allemands* (Paris, 1915), 27–29.

Unlike in France, it is only after the political decision was taken that books and articles began appearing to justify the new policy.⁶⁰ The literature became more strident as Germany's naval and imperial ambitions grew and its relations with Great Britain deteriorated. In Germany, in contrast to France and Britain, the official mind usually preceded public opinion. On the subject of cables, as on so many others, German ideas at that time consisted of a mixture of feelings of persecution and self-assertion. Richard Hennig, a frequent writer on communications questions, complained:

It was England itself which, in the nineties, suddenly sounded a fanfare before an unsuspecting world and repeatedly used its almost total power over cables for national-political purposes in a manner that was extraordinarily painful for the other European peoples. . . . A war between England and another European colonial power in the nineties would have been a . . . gruesome game of cat and mouse.⁶¹

And Thomas Lenschau asserted: "It is all the more important for us that Germany recently also entered into the ranks of states with their own substantial cable networks."⁶²

Germany's first goal was to cross the Atlantic. Felten und Guillaume made a deal with the Commercial Cable Company: In exchange for landing rights in the United States and access to the Postal Telegraph Company's American network, the DAT would share the German-American traffic with Commercial Cable, avoiding their mutual enemies, the Anglo-American and the Western Union Telegraph companies. The DAT also obtained landing rights in the Azores from the Europe and Azores Telegraph Company in exchange for a contract to TC&M to lay the DAT's cable from Borkum to Horta in the Azores, and from there to Coney Island, New York. Thus, the Germans were able to use to their advantage the fierce competition that divided the rival American and British cartels. The new cable, completed on September 1, 1900, was so successful and received so much traffic that it had to be duplicated two years later. The German Atlantic cable of 1902 was the first to be manufactured and laid by Norddeutsche Seekabelwerke.⁶³

In the North Atlantic, Germany benefitted from the goodwill of Portugal and of the Eastern group—not exactly a liberation from British hegemony. To escape the British in other parts of the world—in particular the Middle East, the Western Pacific, and the South Atlantic—Germany needed allies.

The first objective under the new telegraph policy was the Ottoman Empire,

where Germany had growing interests. Communications with Turkey went by one of two ways, neither of them efficient or secure: by Russian land lines to Odessa, then by British cable to Constantinople; or across Austria, Serbia, and Bulgaria. In July 1899 Felten und Guilleaume founded the *Osteuropäische Telegraphengesellschaft* and quickly obtained the right to lay a land line across Hungary and Rumania. For the next five years, it sought landing rights for a cable from Constantza in Rumania to Constantinople, but encountered Turkish procrastination and British intrigues, for this was a part of the world where the British did not relish German interference. Only by threatening to interrupt the Indo-European Telegraph Company lines across Germany in 1905 did the Germans obtain the landing rights they wanted.⁶⁴

But this was only a first step, as observers realized. In the 1890s, Germany had obtained railway concessions from Constantinople to Ankara and on to Bagdad, which the British and Russians found most upsetting. It was clear that Germany also aimed to run a telegraph line along the railway and on to the Persian Gulf. In 1903 Thomas Lenschau wrote: "Perhaps after the completion of the Bagdad Railway the opportunity will arise to continue the connection to the Persian Gulf and thereby obtain a shorter path to India."⁶⁵

In 1908, Germany planned a cable from Basra on the Persian Gulf to Goa and on to Sumatra, but neither Portugal nor Turkey granted the necessary concession. Since their refusal was partly inspired by a fear of British displeasure, the German government approached Foreign Secretary Sir Edward Grey in 1913, asking for "a declaration from His Majesty's Government that they will not oppose in principle the landing of a cable at such a point," that is, the Persian Gulf. The response of the British was that "Sir E. Grey is advised that Germany has no great commercial need for a cable such as that described and he is accordingly led to the conclusion that the object of their request is political." Thus was Germany blocked from the Indian Ocean.⁶⁶

Germany was more successful in the Far East and the Pacific, even though it arrived there late. In 1898 it acquired Tsingtao and Kiaochow Bay in China; in 1899 it bought the Carolinas, Marianas, and Palao from Spain; and in 1900 it obtained part of Samoa. Connections between Tsingtao and Shanghai through Chinese land lines, unsatisfactory in the best of times, were interrupted during the Boxer Rebellion of 1900. The German government therefore had the Great Northern Company lay a cable from Tsingtao to Chefoo and Shanghai, without Chinese authorization.⁶⁷ France, Britain, Russia, and Japan were doing similar and worse things to China at the time.

The new cable joined Tsingtao to Germany through either British or Russian telegraphs; at the time there was no choice. In the Pacific, however, Germany could look for assistance to the Americans and the Dutch.⁶⁸ Throughout the nineteenth century, the Netherlands had lived in the shadow of Britain, and not without misgivings. According to telegraph historian Jorma Ahvenainen, "concern for the safe transmission of cables was also felt in Dutch government circles and the censorship practiced by the British during the Boer War had left the Dutch considerably ill at ease."⁶⁹ Evidently the Dutch, like the French and Germans, chafed at their dependence on Britain for communication with their main colony, the Netherlands East

Indies. When Emil Guilleaume approached the Dutch Telegraph Administration in 1900 about a joint cable project in the Pacific, he found a warm reception, culminating in a treaty signed in Berlin on July 24, 1901. Its purpose was to connect the Dutch East Indies and the German possessions in the Pacific to an eventual American Pacific cable.

When the American cable reached Manila in July 1903, the Dutch and Germans were ready. On July 19, Felten und Guilleaume founded a new subsidiary, the *Deutsch-Niederländische Telegraphengesellschaft*. The next year their cable-manufacturing subsidiary, the *Norddeutsche Seekabelwerke*, laid three cables from the German island of Yap: to Menado on Celebes, connecting with the Netherlands East Indies cable network; to Guam, connecting with the American Pacific cable; and to Shanghai, linked by cable to Kiaochow.

Germany and the Netherlands were now connected to their eastern colonies by non-British cables. This was an expensive form of reassurance, one for which the German government paid 1,525,000 Reichsmark or £76,250 a year, and the Dutch £18,750. Nor was it ever likely to pay for itself, since the agreements of July 26, 1904, with the Eastern Extension, Great Northern, and Commercial Pacific Cable companies stipulated that telegrams via America cost more than via India or Russia. This ensured that business traffic, which constituted nine-tenths of the world's long-distance telegrams, would inevitably avoid the new lines. But no matter, as Richard Hennig explained: "The new lines do not meet any requirement of business nor any requirement of peace, their purpose is much more an eminently strategic one, and just laying them constitutes a move against British cable policy."⁷⁰

The other part of the world that interested Germany before World War I was South America. Thousands of Germans had emigrated to Brazil, Uruguay, Paraguay, and Argentina, 1.5 billion marks were invested there, and hundreds of ships traveled to and from Germany every year, making South America Germany's third trading partner after Europe and North America. Here a direct cable could be justified on commercial grounds. Unfortunately, Brazil and all the Portuguese islands on the way had granted exclusive concessions to the Western and the South American Telegraph companies. But Spain was willing to allow a cable landing at Tenerife in the Canary Islands, and Liberia one at Monrovia. That left Brazil. The concession for a cable from West Africa to Brazil belonged to the South American Telegraph Company, in other words the French government. The *Deutsch-Südamerikanische*, founded in 1908 by Felten und Guilleaume, asked the South American for permission to lay a cable from Monrovia to Pernambuco. On February 4 and March 25, 1910, France and Germany signed the agreements on the South Atlantic cables. The *Norddeutsche Seekabelwerke* laid a Liberia-Brazil cable the following year and a Liberia-Cameroon cable in 1913.⁷¹

The German motives were logical enough. As German trade with South America grew, so did its need to communicate. There was also a political motive associated with the emigration of thousands of Germans to Brazil and Argentina. Charles Lesage, professor at the *Ecole des Sciences Politiques* in Paris and France's foremost expert on the German cable industry, commented: "A manifestation, legitimate up to this point, of German imperialism, the execution of the submarine network is an episode in the general struggle going on for the past twenty years

between Germany and England for predominance in the world."⁷² An arrangement with France was the only way to achieve this goal.

But why did France allow it? Partly, no doubt, for business reasons: In exchange for the concession, Norddeutsche Seekabelwerke completed the French West African network between Conakry and Grand-Bassam for half a million francs, a fraction of its value.⁷³ But this was also a political undertaking, for it was accompanied by an agreement to connect France and Germany by a direct Emden-Brest cable. The French public learned about the cable agreements in October 1911, in the midst of the Second Moroccan Crisis, when a German cable ship sailed into Brest with the end of the cable from Emden. The agreements were therefore misunderstood, not only by the press but even by specialists. Charles Lesage interpreted the French motives in a negative way. It was clear that "Germany had, at times, the support of France in its struggle against the telegraphic dominance of England."⁷⁴ He quoted the German nationalist Max Roscher, who wrote: "The new communication . . . was a public demonstration of the rapprochement between Germany and France in cable matters. . . . [O]n the question of submarine cables, the interests of Germany run parallel to those of France."⁷⁵ Since relations between France and Germany were otherwise hostile, Lesage concluded: "The French Administration at the time entertained with the German telegraph interests relations which were considerably more cordial than those which were painfully being maintained between the Wilhelmstrasse and the Quai d'Orsay."⁷⁶ This situation, he believed, arose because of a flaw in the French political system:

This prolonged disagreement between the general principles of French diplomacy and the procedures of its telegraphic policies come, I believe, from the fact that in this country, each ministry has its own foreign policy: the Ministry of Foreign Affairs has one, the Ministry of Finance has another. . . . The Postal and Telegraph Administration also has, from time to time, a foreign policy; and it so happened, these past few years, that, without being entirely hostile to England, it demonstrated a strong inclination for Germany.⁷⁷

It was, in Lesage's opinion, bad enough that Germany took advantage of the French Telegraph Administration's indulgence in its fight with the British; but this indulgence created a serious misunderstanding in the minds of Chancellor von Bülow and Foreign Minister Bethmann-Hollweg, and "led Germany into error on the true nature of France's relations with England, that is they incited her to believe that the Franco-English Entente was not worth much."⁷⁸

In fact, the cable agreement was not a misguided policy on the part of the Postal and Telegraph Administration, but a byproduct of the fluctuating Moroccan situation. During 1905 and 1906, French and German imperialist ambitions had collided in Morocco. Germany's willingness in 1909 to give up its designs on that country in exchange for a piece of equatorial Africa paved the way for the cable agreements of 1910 that reflected their common interests against the British.⁷⁹ The fact that the two countries were once again at loggerheads over Morocco a year later did not affect the outcome of these agreements.

Conclusion

Between the Boer War and World War I, the world witnessed a second cable boom, comparable in its extent to the boom of the 1870s, but with a different mix of motives and protagonists. This expansion corresponded to a boom in world trade, and many of the new cables—the German Atlantic cables and the American and British Pacific cables in particular—were profitable commercial ventures. Yet in addition to the economic motivation to communicate, every great power felt it needed its own communications for national security reasons. The British, though well represented, no longer had the game to themselves, but had to face competition from the United States, France, and Germany.

Cable rivalry began in the 1880s and accelerated sharply in the 1890s. The increase in world commerce alone would have stimulated the laying of new cables, but the crises of the turn of the century, especially the Boer War, convinced governments of the urgency of having their own cables. Telecommunications was no longer just a business or a public utility but had become one of the pillars of national security. Yet France and Germany could not simply build their own national cable networks parallel to that of Britain, for they lacked the necessary landing licenses, acquired long ago by the British under exclusive concessions. Thus rivalries in telecommunications were not merely a consequence of political and economic rivalries, but contributed to the atmosphere of tension and misunderstanding that was beginning to build up at the turn of the century.

Since that time, technologies of communication have come and gone, but their role in national security, hence in international insecurity, has not diminished.

Notes

1. On the Pacific cable, see George Johnson, ed., *The All-Red Line: The Annals and Aims of the Pacific Cable Project* (Ottawa, 1903).
2. J. Henniker Heaton, "The Postal and Telegraphic Communications of the Empire," *Proceedings of the Royal Colonial Institute* 19 (1887-1888), 171-221, and "Imperial Telegraph System: Cabling to India and Australia," *Contemporary Review* 63 (1893), 537-49. See also Robert J. Cain, "Telegraph Cables in the British Empire 1850-1900," (Ph.D. dissertation, Duke University, 1971), 232-40.
3. Letter from French Consul General Biard in Sydney to Minister of Foreign Affairs Develle, November 22, 1893, and note to Secrétaire d'Etat des Colonies, December 13, 1893, in Archives Nationales Section Outre-Mer (Paris) [henceforth ANSOM] Affaires Politiques 1554 dossier 10: "Création du câble transpacifique, 1892-94."
4. French chargé d'affaires in London to Foreign Minister Develle, October 23, 1893, and Develle to Colonial Minister Boulanger, April 3, 1894, *ibid.*
5. Quoted in Hugh Barty-King, *Girdle Round the Earth: The Story of Cable and Wireless and its Predecessors to Mark the Group's Jubilee, 1929-1979* (London, 1979), 93.
6. Audley Coote to Colonial Office, May 13, 1893, *ibid.*
7. Memorandum from Treasury to Colonial Office, February 3, 1897, in Public Record Office (Kew) [henceforth PRO], CO 42/850, cited in Cain, 257-58.
8. Cain, 255-56.